

**EXPLAINABLE FEATURE DRIFT DETECTION FOR MLOPS
MONITORING**

NUSRAT BEGUM

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Thesis
entitled
**EXPLAINABLE FEATURE DRIFT DETECTION FOR MLOPS
MONITORING**

.....
Ms. Nusrat Begum
Candidate

.....
Asst. Prof. Thanapon Noraset,
Ph.D. (Computer Science)
Major advisor

.....
Assoc. Prof. Suppawong Tuarob,
Ph.D. (Computer Science)
Co-advisor

.....
Assistant Professor Dr. Warin
Krityakiarana
Deputy Dean for Planning and Quality
Assurance
Acting as Dean of the Faculty of Graduate
Studies, Mahidol University
Dean
Faculty of Graduate Studies
Mahidol University

.....
Assoc. Prof. Boonsit Yimwadsana,
Ph.D. (Electrical Engineering)
Program Director
Master of Science Programme
in Computer Science
Faculty of Information and
Communication Technology
Mahidol University

Thesis
entitled
**EXPLAINABLE FEATURE DRIFT DETECTION FOR MLOPS
MONITORING**

was submitted to the Faculty of Graduate Studies, Mahidol University
for the degree of Master of Science (Computer Science)

on
February 2026

.....
Ms. Nusrat Begum
Candidate

.....
Assistant Professor Dr. Akara Supratak,
Ph.D. (Computer Science)
Chair

.....
Asst. Prof. Thanapon Noraset,
Ph.D. (Computer Science)
Member

.....
Assoc. Prof. Suppawong Tuarob,
Ph.D. (Computer Science)
Member

.....
Assistant Professor Dr. Warin
Krityakiarana
Deputy Dean for Planning and Quality
Assurance
Acting as Dean of the Faculty of Graduate
Studies, Mahidol University
Dean
Faculty of Graduate Studies
Mahidol University

.....
Dr. Pattanasak Mongkolwat,
Ph.D. (Computer Science)
Dean
Faculty of Information and
Communication Technology
Mahidol University

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Nusrat Begum

EXPLAINABLE FEATURE DRIFT DETECTION FOR MLOPS MONITORING

NUSRAT BEGUM 6638009 ITCS/M

M.Sc. (COMPUTER SCIENCE)

THESIS ADVISORY COMMITTEE: THANAPON NORASET, Ph.D., SUPPAWONG
TUAROB, Ph.D.

ABSTRACT

Machine learning models in production suffer silent performance degradation from distribution drift. Existing unsupervised detectors signal *that* drift occurred but not *which* features drifted or *what* corrective action to take. This thesis proposes Explainable Adversarial Drift Detection (EADD), integrating discriminative classification with an entropy-based Prediction Uncertainty Index for early warning, adaptive sequential permutation testing, SHAP interaction-aware drift localization, and a novel Drift Severity Index for automated remediation prescriptions. Evaluated against seven state-of-the-art detectors across synthetic and real-world benchmarks, EADD uniquely achieved complete drift-type coverage with near-zero false alarms while providing feature-level root-cause attribution ($P@k = R@k = 1.0$). The adaptive permutation test reduced computational overhead by up to 70% compared to fixed-budget approaches while maintaining statistical rigor.

IMPLICATION OF THE THESIS

**KEY WORDS : DRIFT DETECTION / ADVERSARIAL VALIDATION /
COVARIATE DRIFT / MLOPS / DISTRIBUTION SHIFT /
STREAMING DATA**

88 pages

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CHAPTER I

INTRODUCTION

1.1 Motivation

Machine learning models deployed in production environments operate under the assumption that future data will resemble training data. This assumption is routinely violated: the statistical properties of input features, label distributions, or the conditional relationship between inputs and outputs evolve over time—a phenomenon collectively termed *distribution drift* (Lu et al., 2018; Gama et al., 2014). Left undetected, drift degrades model performance silently, accumulating technical debt and operational risk (Sculley et al., 2015; Bayram et al., 2022). Widmer and Kubat (1996) established that early detection of such changes is essential for maintaining reliable inference in non-stationary environments.

The practical consequences are well documented. Vela et al. (2022) demonstrated that clinical prediction models exhibit measurable accuracy degradation within months of deployment due to temporal distribution shift. Nestor et al. (2019) corroborated this finding, showing that models for common clinical tasks failed to generalise across temporally shifted patient cohorts within the same hospital system. In high-stakes domains, a binary “drift detected” alarm is insufficient; practitioners require immediate root-cause diagnostics to determine whether degradation stems from specific feature pipelines, population-level distributional shifts, or systemic data quality failures (Sculley et al., 2015).

The drift detection landscape has matured considerably. Statistical methods employ hypothesis tests such as the Kolmogorov–Smirnov test or Maximum Mean Discrepancy to compare reference and current distributions (Rabanser et al., 2019; Gretton et al., 2012). Adaptive windowing methods like ADWIN dynamically adjust historical baselines (Bifet and Gavalda, 2007). Supervised error-based detectors (DDM, EDDM) monitor model error rates but require ground-truth labels (Gama et al., 2004;

Baena-García et al., 2006). Prediction-based methods track model outputs but depend on calibration quality (Sethi and Kantardzic, 2017; Guo et al., 2017). Despite this diversity, fundamental challenges persist: univariate statistical tests suffer from false alarm storms in high-dimensional settings; supervised methods cannot operate when labels are delayed or unavailable; and no existing method provides integrated root-cause diagnostics (Lu et al., 2018; Webb et al., 2016).

The discriminative paradigm—recasting distribution comparison as a classification problem—offers a principled alternative. Lopez-Paz and Oquab (2017) formalised this as the *classifier two-sample test* (C2ST): if an auxiliary classifier can reliably separate reference from current data based solely on features, the distributions differ significantly. Gözüaık et al. (2019) adapted this principle for streaming drift detection as the Discriminative Drift Detector (D3), which Lukats et al. (2025) identified as the most robust unsupervised detector in their comprehensive benchmark. However, D3 and all existing discriminative implementations share three critical limitations: (1) they provide only binary detection signals without feature-level attribution, (2) they rely on fixed thresholds lacking statistical justification, and (3) they discard the trained discriminative model without extracting diagnostic value from it.

Recent work has begun addressing the explainability gap. Turrini et al. (2025) proposed feature-interaction aware hypothesis testing, and Hinder et al. (2024c) formalised feature-wise drift notions. Pandeva et al. (2024) demonstrated that e-values can replace p-values in classifier two-sample tests for anytime-valid sequential monitoring. Feng et al. (2024) introduced hierarchical decomposition of performance discrepancies with nonparametric guarantees. However, no existing framework integrates statistically rigorous multivariate detection with quantitative feature-level and interaction-level attribution alongside automated remediation prescriptions in a unified streaming pipeline.

1.2 Objectives

The primary objective of this thesis is to develop **Explainable Adversarial Drift Detection (EADD)**, a framework that advances the discriminative drift detection paradigm through four methodological contributions—uncertainty-aware monitoring,

adaptive sequential permutation testing, interaction-aware feature attribution, and a novel drift severity metric—integrated into a unified streaming pipeline capable of detecting, diagnosing, and prescribing remediation for distribution drift. Specifically, this work aims to:

1. Extend the discriminative drift detection approach of D3 (Gözüaçık et al., 2019) with a non-linear classifier (LightGBM) and **reservoir sampling** for representative reference windows, broadening detection coverage to gradual and incremental drift patterns where linear classifiers fail.
2. Introduce the **Prediction Uncertainty Index (PUI)**, a novel early-warning metric derived from the entropy of the adversarial discriminator, capable of flagging subtle distributional instabilities before they pass statistical significance thresholds.
3. Develop an **adaptive sequential permutation test (ASPT)** that calibrates the number of permutations based on accumulated evidence strength, reducing computational overhead by up to 70% compared to fixed-budget approaches while maintaining Type I error control below the nominal level (Gandy, 2009).
4. Propose **SHAP interaction-aware drift localization** that extends beyond marginal feature attribution to identify correlated feature drift through SHAP interaction values (Lundberg et al., 2020), addressing the limitation that marginal SHAP values can miss joint distributional shifts in correlated feature subspaces.
5. Introduce the **Drift Severity Index (DSI)**, a novel composite metric that combines discriminative power (AUC) with attribution concentration (normalised SHAP entropy) to produce a continuous, theoretically motivated drift severity score, replacing the heuristic thresholds used in prior work.
6. Systematically benchmark EADD against seven state-of-the-art unsupervised drift detectors on real-world datasets from the Lukats et al. (2025) benchmark, evaluating detection accuracy, false alarm control, and the practical value of explainability.

1.3 Expected Contributions

This thesis makes the following contributions:

1. **Prediction Uncertainty Index (PUI):** A novel uncertainty-aware monitoring signal that tracks the entropy of the adversarial discriminator’s predictions. PUI provides a continuous measure of distributional instability, serving as an “early warning” for subtle shifts that have not yet reached statistical significance ($p < 0.01$) but indicate increasing system entropy.
2. **Adaptive Sequential Permutation Testing (ASPT):** A novel early-stopping permutation test for the classifier two-sample test that monitors evidence accumulation and terminates when statistical significance is established or refuted, reducing computational cost by up to 70% compared to fixed-budget permutation testing while preserving Type I error guarantees (Gandy, 2009).
3. **SHAP Interaction-Aware Drift Localization:** Extension of post-hoc feature attribution from marginal SHAP values to SHAP interaction values (Lundberg et al., 2020), enabling detection and diagnosis of correlated feature drift—a common pattern in production data where upstream pipeline failures simultaneously affect multiple related features.
4. **Drift Severity Index (DSI):** A novel composite metric defined as $DSI = AUC \times (1 - H_{\text{norm}}(\mathbf{s}))$, where H_{norm} is the normalised Shannon entropy of the SHAP contribution vector $\mathbf{s}$. DSI replaces heuristic thresholds with a theoretically motivated severity score, enabling severity-proportional remediation responses.
5. **Comprehensive Benchmark:** Systematic comparison of EADD against seven state-of-the-art unsupervised detectors on 11 real-world benchmark streams and four synthetic drift types, demonstrating that methodological innovations in uncertainty monitoring and statistical testing improve detection coverage and false alarm control.

CHAPTER II

LITERATURE REVIEW

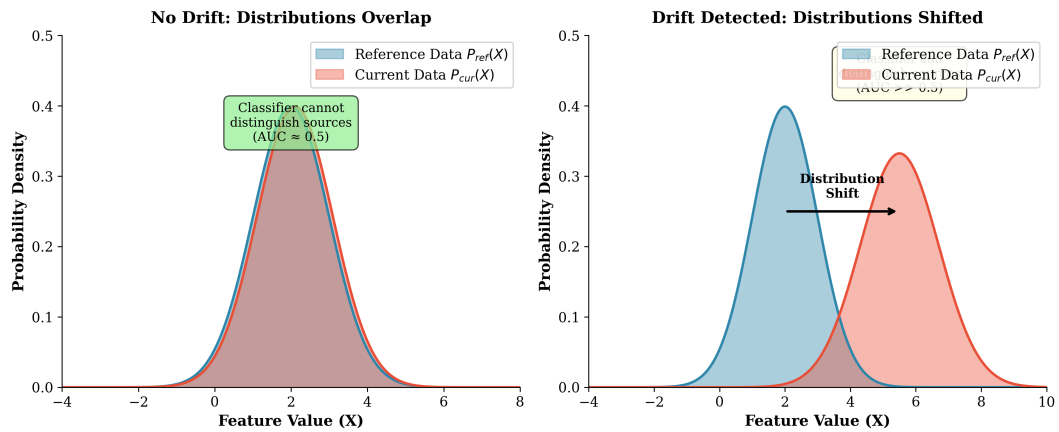


Figure 2.1: Comparison of no drift versus drift scenarios. Left: When distributions overlap, the adversarial classifier cannot distinguish data sources ($AUC \approx 0.5$). Right: When drift occurs, distributions shift and the classifier can distinguish sources ($AUC >> 0.5$).

2.1 Overview

Drift detection addresses the challenge that arises when the statistical characteristics of data evolve over time in a manner that impacts the validity of a machine learning model (Lu et al., 2018; Gama et al., 2014). According to Bayram et al. (2022), when users, environments, data sources, or the relationships between labels and features change, model accuracy can diminish. Gama et al. (2014) and Sculley et al. (2015) emphasize that drift detection identifies such modifications early, allowing practitioners to adapt, retrain, or update the model accordingly. A robust understanding of drift types and detection methods is essential for designing effective monitoring systems.

2.2 Historical Evolution of Drift Detection

The study of distribution drift has evolved from foundational statistical concepts to integrated AI governance over roughly seven decades. Table 2.1 summarises the major eras.

Table 2.1: Historical evolution of drift detection research.

Era		Key Developments
Early Foundations (1950s–1980s)		Statistical process control and sequential analysis laid the groundwork. Page (1954) introduced the cumulative-sum (CUSUM) change-point test, which remains foundational for sequential change detection. Reservoir sampling (Vitter, 1985) enabled representative subsampling of data streams.
Identification Era (1998–2004)		The term “concept drift” gained formal academic recognition (Widmer and Kubat, 1996). Gama et al. (2004) introduced the Drift Detection Method (DDM), using PAC-learning error-rate bounds to trigger alerts—establishing the supervised error-monitoring paradigm.
Windowing & Statistical Tests (2007–2015)		Bifet and Gavalda (2007) proposed ADWIN (Adaptive Windowing), enabling drift detection without fixed window sizes by applying statistical tests over dynamically adjusted sub-windows. Non-parametric tests (KS, MMD) became standard for comparing reference and current distributions (Rabanser et al., 2019; Gretton et al., 2012).
Deep Learning & High-Dimensionality (2018–2023)		As models moved to unstructured data, detection shifted toward monitoring internal latent representations (Rabanser et al., 2019; Liu et al., 2020). Gözüaık et al. (2019) introduced D3, demonstrating that discriminative classifiers could serve as powerful multivariate drift detectors. Lopez-Paz and Oquab (2017) formalised the classifier two-sample test (C2ST), providing theoretical backing for this approach.
Governance & Explainability Era (2024–2026)		Drift detection became embedded in AI governance frameworks. The focus shifted toward interpretable, statistically rigorous detection: TRIPODD (Turrini et al., 2025) introduced feature-interaction hypothesis testing; E-C2ST (Pandeva et al., 2024) enabled anytime-valid sequential testing via e-values; HDPD (Feng et al., 2024) decomposed performance gaps into $P(X)$ and $P(y X)$ components with attributions. Cloud platforms standardised SHAP-based feature attribution monitoring (Taly and Sato, 2021; Amazon Web Services, 2025). Hinder et al. (2024a) exposed vulnerabilities in discriminative detectors to adversarial evasion.

This progression—from reactive error monitoring to proactive, explainable distribution analysis—provides the historical context for the present work, which integrates discriminative detection with SHAP-based attribution in the streaming setting.

2.3 Types of Drift

Drift can be categorized along two main dimensions: by the nature of the change (what component of the data-generating process is affected) and by the temporal pattern of the change (how the drift manifests over time) (Lu et al., 2018; Gama et al., 2014).

2.3.1 Drift by Nature of Change

Based on which component of the joint distribution $P(X, y)$ changes, the literature (Lu et al., 2018; Gama et al., 2014; Tsymbal, 2004) classifies drift into the following types:

Concept Drift (Real Drift): The conditional distribution $P(y|X)$ changes over time while the feature distribution $P(X)$ may or may not change (Lu et al., 2018; Žliobaitė, 2010). This represents a change in the underlying relationship between inputs and outputs—the “concept” the model has learned becomes invalid. For example, the factors that determine creditworthiness may evolve due to economic shifts, even if the applicant characteristics (features) remain similar (Gama et al., 2014). Concept drift directly degrades model performance because the learned decision boundaries no longer reflect the true input-output relationship (Lu et al., 2018).

Covariate Drift (Feature Drift / Virtual Drift): The marginal distribution of input features $P(X)$ changes over time while the conditional relationship $P(y|X)$ remains stable (Tsymbal, 2004; Shimodaira, 2000). The ground-truth concept does not change, but the model may encounter inputs outside its training distribution. For example, a fraud detection model trained on transaction patterns from one demographic may receive data from a different user population, even though the definition of fraud remains unchanged (Tsymbal, 2004). Covariate drift can degrade model performance because the model is exposed to feature regions where it may not generalize well (Shimodaira, 2000).

Prior Probability Drift (Label Shift): The marginal distribution of labels $P(y)$ changes over time while $P(X|y)$ remains stable (Tsymbal, 2004). This occurs when the prevalence of different classes changes—for instance, a seasonal increase in a particular disease affecting the class balance in medical diagnosis. This is sometimes called class imbalance shift (Lu et al., 2018).

Note on Terminology: The literature uses varied terminology (Lu et al., 2018). The term “concept drift” is sometimes used as an umbrella term for any distribution change affecting model performance (Gama et al., 2014), while some reserve it specifically for changes in $P(y|X)$ (Tsymbal, 2004). This thesis focuses primarily on covariate (feature) drift detection, as it enables early warning without requiring labels.

2.3.2 Drift by Temporal Pattern

Based on how drift manifests over time, the literature (Lu et al., 2018; Gama et al., 2014; Žliobaitė, 2010) categorizes drift as follows:

- **Abrupt (Sudden) Drift:** A rapid, discrete change where the data distribution shifts quickly from one state to another within a short time period (Gama et al., 2014).
- **Gradual Drift:** A new distribution progressively replacing the old one over an extended duration, with samples from both distributions interleaved during the transition period (Lu et al., 2018).
- **Incremental Drift:** Small but continuous changes accumulating over time, eventually resulting in a significant distributional shift (Žliobaitė, 2010).
- **Recurring (Seasonal) Drift:** Previously observed distributions reappearing periodically due to cyclical patterns, such as seasonal effects in retail or recurring user behavior patterns (Lu et al., 2018; Gama et al., 2014).

Understanding drift patterns is important for detector design: abrupt drift requires rapid response, gradual drift requires sustained sensitivity, and recurring drift benefits from mechanisms that can recognize previously seen distributions (Gama et al., 2014).

2.3.3 Summary Taxonomy

Table 2.2 consolidates the drift classification used throughout this thesis.

Table 2.2: Taxonomy of drift types referenced in this thesis.

Category	Type	Description
By Cause (Lu et al., 2018; Gama et al., 2014)	Data / Feature Drift	Change in the input distribution $P(X)$ without changing the relationship to the target.
	Concept Drift	Change in the conditional $P(y X)$, invalidating learned decision boundaries.
	Prior Probability Drift	Change in the label distribution $P(y)$ while $P(X y)$ is stable.
By Pattern (Gama et al., 2014; Žliobaitė, 2010)	Sudden (Abrupt)	Rapid, discrete shift between two distinct distributions.
	Gradual	Old concept progressively replaced by a new one over an extended period.
	Incremental	Small, continuous changes accumulating into a significant shift.
	Recurring (Seasonal)	Previously observed distributions reappearing cyclically.

This thesis focuses primarily on **feature (covariate) drift**—changes in $P(X)$ —because it can be detected without ground-truth labels and provides early warning before model performance degrades.

2.4 Classification of Drift Detection Methods

Based on comprehensive surveys of drift detection literature (Lu et al., 2018; Gama et al., 2014; Tsymbal, 2004; Hinder et al., 2024b), drift detection methods can be classified by multiple dimensions. According to Lu et al. (2018), the most fundamental distinction is based on label availability: whether the method requires access to ground-truth labels (supervised) or operates without them (unsupervised).

2.4.1 Supervised Drift Detection Methods

According to Gama et al. (2014) and Gama et al. (2004), supervised drift detection methods require access to ground-truth labels and primarily monitor model performance or prediction errors to detect distributional changes. Gama et al. (2004)

and Baena-García et al. (2006) explain that these methods operate under the assumption that unexpected changes in error characteristics—such as increases in error rate, loss magnitude, variance, or cumulative deviation—are indicative of underlying drift.

Key supervised drift detection algorithms include:

Drift Detection Method (DDM): According to Gama et al. (2004), DDM monitors the online error rate and its standard deviation. When these statistics exceed predefined thresholds derived from binomial distribution bounds, DDM signals a warning or drift state. Gama et al. (2004) note that DDM is effective for abrupt drift but may be slow to detect gradual changes.

Early Drift Detection Method (EDDM): EDDM extends DDM by tracking the distance between consecutive classification errors rather than just the error rate (Baena-García et al., 2006). This modification improves sensitivity to gradual drift by detecting when errors cluster more closely together.

Page–Hinkley Test: This sequential analysis technique uses cumulative sums to detect changes in the mean of a monitored signal (Page, 1954). When cumulative deviation exceeds a threshold, drift is indicated.

Hoeffding Drift Detection Methods (HDDM-A, HDDM-W): These methods use Hoeffding’s inequality to provide statistically grounded drift decisions with theoretical guarantees (Frias-Blanco et al., 2015). HDDM-A detects abrupt changes using moving averages, while HDDM-W is designed for gradual drift using window-based statistics.

Limitations: Supervised methods fundamentally require labeled data, which may be delayed, expensive, or unavailable in many production scenarios (Lu et al., 2018; Sethi and Kantardzic, 2017). They also react to drift after it has already caused performance degradation, rather than detecting distributional change proactively.

2.4.2 Unsupervised Drift Detection Methods

Unsupervised drift detection methods do not require ground-truth labels and instead monitor changes in data distribution, model outputs, or internal representations (Tsymbal, 2004; Hinder et al., 2024b). These methods enable earlier detection of distributional shifts—before performance degradation becomes observable—making them

particularly valuable in production environments with label delays (Sethi and Kantardzic, 2017).

2.4.2.1 Statistical Distribution-Based Methods

Statistical test-based methods detect drift by directly examining changes in data distributions through hypothesis testing or distance-based measures (Rabanser et al., 2019; Raab et al., 2020). A common strategy compares a reference data window against a current data window to test whether they originate from the same distribution (Bifet and Gavaldà, 2007).

Key statistical drift detection methods include:

Adaptive Windowing (ADWIN): ADWIN dynamically maintains a sliding window and automatically shrinks it when statistically significant distributional change is detected (Bifet and Gavaldà, 2007). It provides theoretical guarantees on false positive and false negative rates and adapts its window size based on observed change.

KSWIN: This method applies the Kolmogorov–Smirnov two-sample test over sliding windows to detect changes in one-dimensional feature distributions (Raab et al., 2020). KSWIN is non-parametric and makes minimal assumptions about the data distribution.

Chi-Squared Test-Based Detectors: These use the chi-squared test to compare categorical feature distributions or discretized continuous features between time periods (Lu et al., 2018).

Kullback–Leibler (KL) Divergence and Other Divergence Measures: Methods that quantify the “distance” between probability distributions, including KL divergence, Jensen–Shannon divergence, and Hellinger distance, can all be used to measure distributional shift (Dasu et al., 2006).

Maximum Mean Discrepancy (MMD): MMD is a kernel-based two-sample test that compares distributions by mapping them into a reproducing kernel Hilbert space, capable of capturing complex multivariate distributional differences (Gretton et al., 2012).

Strengths and Limitations: Statistical tests are interpretable, theoretically grounded, and label-free (Rabanser et al., 2019). However, they can be sensitive to noise (causing false alarms), may struggle in high-dimensional spaces due to the curse of dimensionality, and typically test marginal distributions rather than joint distributions

(Lu et al., 2018; Webb et al., 2016).

Comparison of Statistical Methods with Discriminative Classification:

Table 2.3 compares popular univariate statistical methods (PSI, KS) with the multivariate discriminative classification approach used in this thesis.

Table 2.3: Comparison of Drift Detection Approaches

Method	Approach	Strengths	Limitations
PSI (Population Stability Index)	Compares binned distributions of single features; $\text{PSI} > 0.25$ indicates significant drift	Simple, interpretable, industry standard	Univariate only; requires binning; misses feature correlations
KS Test (Kolmogorov-Smirnov)	Non-parametric test comparing cumulative distributions per feature	No distributional assumptions; exact p-values	Univariate only; multiple testing problem with many features
Discriminative Classification	Trains classifier to distinguish old vs. new data; uses AUC as drift signal	Multivariate; captures feature interactions; single unified test	Requires classifier training; no built-in explainability (addressed by EADD)

The key distinction is that PSI and KS are **univariate** methods—they test each feature independently. When monitoring d features, this requires d separate tests, leading to the “false alarm storm” problem where the probability of at least one false positive increases with d . Discriminative classification is inherently **multivariate**, testing the joint distribution $P(X_1, X_2, \dots, X_d)$ in a single unified check.

2.4.2.2 Model-Based Discriminative Methods

According to Pan et al. (2020), Lopez-Paz and Oquab (2017), and Sethi and Kantardzic (2017), discriminative drift detection methods train an auxiliary model to distinguish between reference and current data distributions. Lopez-Paz and Oquab (2017) explain that the core idea is to frame drift detection as a classification problem: if a classifier can accurately predict whether each instance originates from the reference or current distribution, the distributions must be distinguishably different.

Key approaches include:

Domain Classifier / Discriminative Distribution Testing: According to Pan et al. (2020) and Qian and Rao (2021), this approach trains a binary classifier (e.g., logistic

regression, random forest, gradient boosting) to discriminate samples from reference vs. current windows. Pan et al. (2020) note that high discriminative performance (measured by accuracy, AUC, or similar metrics) indicates significant distributional shift. The technique is sometimes called “adversarial validation” in industry practice (Pan et al., 2020).

Discriminative Drift Detector (D3): Gözüa ık et al. (2019) developed this unsupervised method that trains a classifier (logistic regression) to distinguish between old and new data windows, using AUC as the drift signal. D3 has demonstrated strong performance in recent benchmarks (Lukats et al., 2025) and represents the state-of-the-art in discriminative drift detection.

Margin Density Drift Detection (MD3): Sethi and Kantardzic (2017) proposed an alternative approach that monitors margin density—the proportion of samples near the decision boundary—to detect drift without requiring true labels.

Margin-Based and Density-Based Discriminators: According to Ditzler and Polikar (2011), these methods monitor changes in classifier margin densities or decision boundary proximity to detect distributional shifts.

Strengths: Discriminative methods can capture high-dimensional, non-linear distributional shifts that marginal univariate tests may miss (Lopez-Paz and Oquab, 2017). They require no labels for the drift detection task itself (only pseudo-labels indicating data source) and naturally aggregate information across all features (Pan et al., 2020).

Limitations: These methods require training a classifier at each monitoring step, which has computational cost; performance depends on classifier choice and hyper-parameters; and interpretation of drift signals may be less intuitive than statistical test p-values (Sethi and Kantardzic, 2017).

2.4.2.3 Representation-Based Methods

Representation-based methods detect drift by monitoring changes in learned representations or embeddings rather than raw feature distributions (Rabanser et al., 2019; Liu et al., 2020):

Embedding Distribution Shift: Computing statistical distances (cosine similarity, Wasserstein distance, MMD) between reference and current embedding distributions derived from neural network intermediate layers (Rabanser et al., 2019).

Deep Learning Representation Monitoring: This approach tracks stability of learned representations across time, leveraging insights from domain adaptation and transfer learning literature (Long et al., 2016).

These methods are particularly relevant for deep learning systems where raw feature distributions may not reflect semantically meaningful changes (Liu et al., 2020).

2.4.2.4 Prediction-Based / Output-Based Methods

Prediction-based methods detect drift by monitoring changes in model outputs over time without requiring evaluation of prediction correctness (Sethi and Kantardzic, 2017; Gözüaık et al., 2019):

Prediction Distribution Shift: Monitoring changes in the distribution of model predictions (class probabilities, regression outputs) over time (Gözüaık et al., 2019).

Uncertainty / Entropy Monitoring: Tracking prediction entropy or uncertainty scores under the assumption that rising uncertainty reflects exposure to out-of-distribution data (Guo et al., 2017).

Confidence Monitoring: Tracking changes in model confidence levels and softmax output distributions (Guo et al., 2017).

Limitations: These methods depend on model calibration. If the model is poorly calibrated—which is common for modern deep learning models—changes in confidence may not reliably indicate drift (Guo et al., 2017; Ovadia et al., 2019).

2.4.2.5 State-of-the-Art Unsupervised Detectors (2019–2025)

Recent benchmarking by Lukats et al. (2025) provides a systematic evaluation of fully unsupervised drift detectors on real-world data streams. Their study tested numerous algorithms and identified seven that function reliably on messy, real-world data. This survey represents a critical “reality check” for the field, demonstrating that detecting change without ground-truth labels is possible but fragile.

D3 (Discriminative Drift Detector): According to Gözüaık et al. (2019), D3 trains a logistic regression classifier to distinguish between old and new data windows, using AUC as the drift signal. Lukats et al. (2025) found D3 to be the overall winner—the most robust and accurate across different data types. However, D3 remains a “black box”: it signals *that* drift occurred but provides no insight into *which* features drifted or *why*.

CSDDM (Clustered Statistical Drift Detection Method): According to Hsieh et al. (2021), this method uses PCA for dimensionality reduction followed by K-Means clustering, then applies statistical tests to detect cluster shifts. Lukats et al. (2025) report it performed exceptionally well on the “lift-per-drift” metric, catching drifts with minimal false alarms.

SPLL (Semi-Parametric Log-Likelihood): Kuncheva (2013) developed this approach that fits probability density models to reference data and detects drift when new observations have anomalously low likelihood under the learned distribution. It is effective for sudden drift patterns.

IBDD (Image-Based Drift Detector): According to Souza et al. (2020b), this creative approach converts numerical data streams into images and applies image comparison techniques to detect distributional changes. While highly accurate, Lukats et al. (2025) note it can be biased by its frequent reset strategy.

OCDD (One-Class Drift Detector): Gözüaık and Can (2021) proposed using one-class SVMs to define “normal” data boundaries and flag drift when incoming data falls outside these boundaries. While sound in principle, D3 generally outperformed OCDD in benchmarks (Lukats et al., 2025).

NN-DVI (Nearest Neighbor Density Variation): According to Liu et al. (2018), this method monitors changes in local data density by tracking nearest-neighbor distances. Drift is signaled when density patterns shift significantly. It performs well for gradual density changes but can be sensitive to noise.

UCDD (Unsupervised Concept Drift Detection): Shang et al. (2020) developed a clustering-based approach that assigns pseudo-labels based on cluster membership and monitors boundary shifts. It serves as a reliable baseline in unsupervised drift detection.

BNDM (Bayesian Non-parametric Drift Monitor): This method uses a Pólya tree test to detect distributional changes without parametric assumptions. However, Lukats et al. (2025) report that BNDM struggled with small sample sizes in their benchmark, limiting its practical applicability in streaming scenarios where window sizes are constrained.

UDetect (Univariate Detection): This approach monitors standard deviation

changes using Shewhart control charts. While simple and computationally efficient, Lukats et al. (2025) found it had a high failure rate (45%) in their benchmark, making it unreliable for production deployment.

Industrial Drift Detection Libraries: Beyond research algorithms, several industrial libraries provide production-ready drift detection:

- **Alibi Detect:** Open-source Python library (by Seldon) implementing multivariate drift detection via Kolmogorov–Smirnov tests, MMD, and learned kernel methods.
- **NannyML:** Open-source library providing Confidence-Based Performance Estimation (CBPE) for monitoring model performance without ground-truth labels.
- **Evidently AI:** Open-source ML monitoring framework with built-in statistical drift tests for tabular data.

Gap Identified from Benchmark Analysis: While these libraries and algorithms exist, according to Lukats et al. (2025), they share a common limitation: they typically report only *whether* drift occurred, not *where* or *why*. Statistical libraries often run separate univariate tests for each feature, causing “false alarm storms” when many features are monitored simultaneously. Although Lopez-Paz and Oquab (2017) proposed the classifier two-sample test (C2ST) framework with principled significance testing, existing streaming implementations such as D3 (Gözüaık et al., 2019) use only fixed AUC thresholds and discard the learned classifier without exploiting it for explainability. The field lacks an integrated method that combines a statistically rigorous multivariate check with granular, feature-level explanations of drift sources.

2.4.3 Hybrid and Ensemble Methods

Hybrid approaches integrate multiple sources of information—both supervised and unsupervised signals—to improve drift detection robustness (Gama et al., 2014; Krawczyk et al., 2017):

Context-Aware Drift Detection: Combining model predictions, user context, operational signals, and metadata to compute context-weighted drift scores (Krawczyk

et al., 2017).

Multi-Signal Ensemble Methods: Integrating predictions, embeddings, operational KPIs, and metadata into a unified drift score (Krawczyk et al., 2017).

Drift Detection Ensembles: Combining multiple base detectors using voting, weighting, or meta-learning to reduce both false alarms and missed detections (Krawczyk et al., 2017).

2.5 Limitations of Existing Methods

While the drift detection literature offers diverse approaches, each category has characteristic limitations:

Supervised Methods (DDM, EDDM, HDDM): These require timely labeled data, which is often delayed or unavailable in production (Lu et al., 2018). They detect drift reactively—only after performance degradation has already occurred (Sethi and Kantardzic, 2017).

Statistical Distribution Methods (ADWIN, KSWIN, KL-Divergence): These can be sensitive to noise, leading to false alarms (Rabanser et al., 2019). They may struggle in high-dimensional or correlated feature spaces and typically test marginal distributions, potentially missing joint distribution changes (Lu et al., 2018; Webb et al., 2016).

Prediction/Uncertainty-Based Methods: These rely on model calibration. Modern ML models are often poorly calibrated, especially under distribution shift, causing these signals to be unreliable (Guo et al., 2017; Ovadia et al., 2019).

Context-Aware Methods: These require pre-specification of relevant context variables, which may be incomplete, noisy, or themselves drift over time (Krawczyk et al., 2017). Misspecified context can cause missed detections or spurious alarms.

Fundamental Gap: Critically, most existing methods do not directly test whether new data is statistically distinguishable from historical data in the original feature space (Lopez-Paz and Oquab, 2017). Discriminative approaches offer a natural, model-agnostic signal of drift that does not depend on predefined contexts, labels, or calibrated

uncertainties—yet this perspective has received limited systematic attention in the drift detection literature (Pan et al., 2020).

This thesis aims to bridge this gap by establishing discriminative classification as a viable, general-purpose drift detection technique and extending it with explainability through rigorous empirical evaluation and practical methodology development.

2.6 Drift Mitigation and Adaptation Strategies

Detecting drift is only the first step; production systems must also respond to it. The literature describes five principal mitigation strategies, each with distinct trade-offs (Lu et al., 2018; Gama et al., 2014; Krawczyk et al., 2017):

1. **Periodic Retraining:** The model is retrained on the latest data at fixed intervals (e.g., weekly or monthly) or triggered by drift alerts. This is the most common industrial approach because of its operational simplicity, but fixed schedules may retrain unnecessarily or react too slowly (Gama et al., 2014).
2. **Weighted Retraining:** Old and new data are combined for retraining, but recent examples receive higher weights. This preserves long-term knowledge while prioritising current patterns, and is particularly effective for gradual drift (Krawczyk et al., 2017).
3. **Ensemble Methods:** A library of models—e.g., one per season or data regime—is maintained, and predictions are combined via voting, stacking, or dynamic weighting. When drift is detected, poorly performing members are discarded and new ones trained on the current distribution (Krawczyk et al., 2017).
4. **Online / Incremental Learning:** The model is updated continuously as each new data point (or mini-batch) arrives, making it naturally adaptive to gradual and incremental drift. This is common in streaming applications such as fraud detection and recommendation systems (Gama et al., 2014; Ditzler et al., 2015).

5. **Human-in-the-Loop:** Uncertain or drifted samples are routed to human experts for re-labelling, which feeds into the next training iteration. This strategy is essential in domains where errors carry high cost (e.g., health-care, autonomous driving) and labelling is expensive (Lu et al., 2018).

Gap in Mitigation Strategies: Existing mitigation approaches assume that the practitioner has already diagnosed the drift and selected the correct response. In practice, the choice between a targeted feature-pipeline fix, partial retraining on a subset of features, and full retraining depends on *which* features drifted and by *how much*—diagnostic information that current detectors do not provide. EADD’s automated prescription system (Chapter 3) addresses this gap by mapping SHAP-based attribution patterns directly to mitigation actions.

2.7 Industry Applications of Drift Detection

Drift detection and mitigation are now deployed across multiple high-stakes domains:

Finance: Credit scoring models are particularly susceptible to “segment drift,” where certain demographic groups’ financial behaviours shift due to economic conditions. Qian and Rao (2021) demonstrated discriminative detection for credit scoring, filtering training samples to maintain alignment between training and deployment distributions. Population Stability Index (PSI) remains the industry-standard univariate monitor for regulatory compliance.

Healthcare: Clinical prediction models degrade when patient populations change or when data acquisition technology evolves. Nestor et al. (2019) showed that ML models for common clinical tasks failed to generalise across temporally shifted cohorts within the same hospital. Vela et al. (2022) documented measurable accuracy losses within months of deployment, underscoring the need for continuous monitoring in medical AI.

Cybersecurity: Malware detection models face adversarial concept drift as attack patterns evolve to evade existing detectors. Static models become obsolete rapidly,

and drift-triggered retraining is critical for maintaining detection rates against emerging threats (Ditzler et al., 2015).

E-Commerce and Recommendations: Recommendation engines must adapt to seasonal demand spikes (e.g., Black Friday, holiday periods) and evolving user preferences. Recurring drift patterns are common, and ensemble-based switching between seasonal models is a practical response (Gama et al., 2014; Krawczyk et al., 2017).

Cloud MLOps Platforms: As described in Section 2.8.2, Google Vertex AI (Taly and Sato, 2021) and AWS SageMaker Clarify (Amazon Web Services, 2025) have standardised feature attribution monitoring in production, signalling industry consensus that drift diagnostics must move beyond binary alarms toward explainable, feature-level attribution.

2.8 Related Work: Discriminative and Explainable Approaches

2.8.1 Discriminative Distribution Testing

Discriminative approaches to distribution comparison reframe the problem as a binary classification task. According to Pan et al. (2020) and Qian and Rao (2021), a classifier is trained to predict whether each instance originates from a reference distribution or a target distribution. Lopez-Paz and Oquab (2017) formalized this as the *classifier two-sample test* (C2ST), showing that high discriminative performance indicates statistical separability—hence significant distributional shift—while near-random performance suggests the distributions are largely indistinguishable.

Lopez-Paz and Oquab (2017) demonstrated that this discriminative approach can detect complex multivariate shifts that marginal statistical tests miss, and provided theoretical foundations for significance calibration via permutation testing. The approach is label-independent, model-agnostic, and makes no assumptions about which features are important.

2.8.2 Industrial Applications of Discriminative Detection

Uber's MaLTA System: Pan et al. (2020) applied discriminative distribution testing at Uber for their MaLTA (Machine Learning Targeting Automation) user targeting system. They trained a classifier to distinguish historical training data from newly observed targeting data. When the classifier exhibited strong separability, it signaled train-deployment mismatch, prompting feature filtering and retraining adjustments before deployment.

Credit Scoring: Qian and Rao (2021) applied discriminative testing to manage dataset shift in credit scoring, using the classifier's predictions to filter training samples—keeping only those most similar to the current deployment distribution.

Image Classification and Operations: Additional applications include quantifying dataset complexity in image classification Renza and Moya-Albor (2024) and detecting when historical data is no longer representative in manufacturing/operations settings Senoner et al. (2023).

Feature Attribution Monitoring at Google: In an industry report, Taly and Sato (2021) described how Google's Vertex AI platform uses feature attribution monitoring to detect model degradation in production. Their system computes feature attribution baselines during model training and continuously compares production attributions against these baselines, alerting engineers with diagnostic information about which features changed. According to the report, this approach detected a service degradation incident that traditional input-distribution monitoring missed, because the drift manifested in feature importance patterns rather than in raw input statistics. While this is industry-reported evidence rather than peer-reviewed research, the deployment at Google scale suggests that feature-level attribution analysis can provide superior drift diagnostics compared to aggregate distribution tests.

Cloud Industry Standardization: The necessity of feature attribution monitoring is rapidly becoming an industry standard across major cloud providers. As with Google's Vertex AI, Amazon Web Services (AWS) has integrated "Feature Attribution Drift" monitoring into Amazon SageMaker Clarify (Amazon Web Services, 2025). SageMaker automatically calculates a SHAP baseline during training and actively monitors production data for deviations in these attribution scores. This cross-platform adoption

by leading cloud providers indicates that drift detection is increasingly moving toward continuous, explainable feature attribution rather than basic distribution tracking.

2.8.3 Recent Advances in Drift Detection (2024–2026)

The drift detection landscape has evolved rapidly along three axes: statistical rigour, explainability, and computational efficiency.

Statistical Testing Advances. Pandeva et al. (2024) proposed E-C2ST, which replaces p-values with e-values in classifier two-sample tests, enabling anytime-valid sequential monitoring without multiple testing penalties. This represents a strictly superior alternative to fixed-budget permutation testing for sequential settings. Wan et al. (2024) introduced Maximum Concept Discrepancy for online drift detection using contrastive learning, offering an alternative to the discriminative paradigm.

Explainability and Attribution. Turrini et al. (2025) proposed feature-interaction aware hypothesis testing with formal power guarantees, enabling identification of which specific feature interactions drive detected drift. Feng et al. (2024) introduced a hierarchical nonparametric decomposition at NeurIPS 2024 that separates covariate shift ($P(X)$) from concept shift ($P(y|X)$) with per-feature Shapley attributions and confidence intervals. Edakunni et al. (2024) decomposed performance drift into virtual and real drift components using distributional Shapley values. Decker et al. (2024) combined optimal transport with Shapley values to attribute performance degradation to specific feature shifts. Lyu et al. (2025) proposed GAM-based concept shift characterisation with knockoff-based false discovery rate control, achieving inherently interpretable shift explanations. Komnick et al. (2025) argued that only causal (not correlational) drift explanations are truly actionable, introducing causal attribution methods for drift remediation.

Formal Frameworks. Hinder et al. (2024c) formalised a rigorous mathematical definition of feature-wise drift, enabling detector-agnostic feature selection for drift localisation. A systematic literature review by Various Authors (2025) confirmed that while the intersection of XAI and drift detection is rapidly growing, most existing approaches adapt static XAI methods to streaming settings without addressing the fundamental challenges of scalability and interaction-level analysis.

Label Efficiency. Abdi et al. (2024) achieved label-efficient semi-supervised detection, and Cerqueira et al. (2022) proposed student–teacher methods for unsupervised detection. Greco et al. (2024) addressed drift in unstructured data through per-label embedding distribution monitoring. Song et al. (2022) introduced autoregressive error monitoring with confidence intervals.

Robustness Concerns. Hinder et al. (2024a) demonstrated that adversarial data streams can be constructed to evade common drift detectors, including discriminative approaches, highlighting the need for robust detection mechanisms.

Key Observation: While these advances have substantially improved individual aspects of drift detection, a critical gap persists: no existing framework integrates (1) computationally efficient, statistically rigorous multivariate detection with (2) interaction-aware feature-level attribution and (3) automated, severity-calibrated remediation prescriptions in a unified streaming pipeline. The closest approaches—TRIPODD for interaction-aware testing, E-C2ST for sequential statistical rigour, and HDPD for attribution decomposition—each address only one dimension of this triad.

2.9 Gap in the Literature

The preceding review reveals a progressive evolution from supervised error monitoring toward unsupervised, interpretable drift detection. Recent advances—TRIPODD’s interaction-aware testing (Turrini et al., 2025), E-C2ST’s sequential e-value testing (Pandeva et al., 2024), HDPD’s hierarchical decomposition (Feng et al., 2024), and CDSeer’s label-efficient detection (Abdi et al., 2024)—have substantially advanced individual aspects. However, five critical gaps persist:

1. **The “Black Box” Problem:** The benchmark-leading D3 (Lukats et al., 2025) and most unsupervised detectors provide binary detection signals without feature-level attribution. While TRIPODD (Turrini et al., 2025) addresses interactions and HDPD (Feng et al., 2024) provides attributions, neither operates as a unified streaming detection-and-diagnosis pipeline.

2. **Marginal Attribution Blindness:** Methods that provide feature-level explanations—including standard SHAP analysis and the XPE framework (Decker et al., 2024)—typically report marginal feature contributions. They miss correlated feature drift where the interaction between features shifts while marginal distributions remain stable (Hinder et al., 2024c). Interaction-level diagnosis remains largely unexplored in the streaming drift detection context.
3. **Computational Inefficiency of Statistical Calibration:** Fixed-budget permutation testing (the standard approach for calibrating classifier two-sample tests (Lopez-Paz and Oquab, 2017)) requires a predetermined number of permutations regardless of evidence strength. When drift is obvious or clearly absent, most permutations are wasted computation. Adaptive stopping rules that maintain Type I error guarantees have been developed for Monte Carlo tests (Gandy, 2009) but have not been applied to discriminative drift detection.
4. **Absence of Continuous Severity Quantification:** Existing detectors produce binary decisions (drift/no-drift) without quantifying drift severity. Production MLOps systems require a continuous severity metric to calibrate response urgency—distinguishing, for example, between a catastrophic sensor failure requiring immediate intervention and a gradual population shift warranting scheduled retraining.
5. **No Automated Remediation Mapping:** Even when drift is detected and partially explained, the decision of *how to respond* remains manual (Bayram et al., 2022; Ditzler et al., 2015). No existing method maps quantitative drift diagnostics to severity-calibrated remediation prescriptions.

This thesis addresses these gaps by proposing **Explainable Adversarial Drift Detection (EADD)**, which makes four specific methodological contributions beyond the established discriminative detection paradigm (Gözüaık et al., 2019; Lopez-Paz and Oquab, 2017): (1) a **Prediction Uncertainty Index (PUI)** that provides an early-warning signal from the entropy of the adversarial discriminator’s predictions, (2) an **adaptive**

sequential permutation test that reduces computational cost while maintaining statistical rigour, (3) **SHAP interaction-aware drift localisation** that captures correlated feature shifts invisible to marginal attribution, and (4) a novel **Drift Severity Index** that provides a continuous, theoretically motivated severity metric enabling automated, proportional remediation prescriptions.

CHAPTER III METHODOLOGY

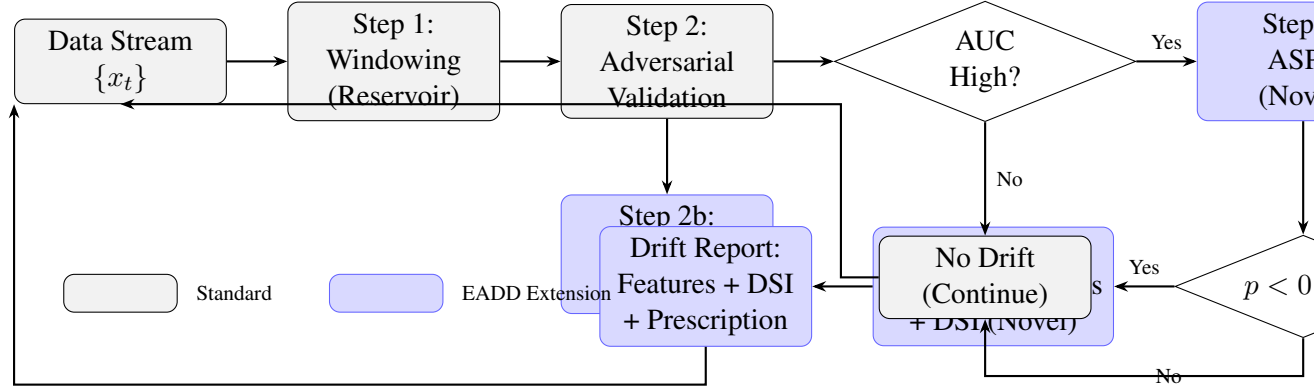


Figure 3.1: Overview of Explainable Adversarial Drift Detection (EADD) methodology. Blue boxes indicate EADD’s four novel extensions beyond D3: PUI early warning (Step 2b), Adaptive Sequential Permutation Test (Step 3), and SHAP Interaction-Aware Attribution with DSI (Step 4). The framework integrates discriminative classification with uncertainty monitoring, statistical rigour, and root-cause analysis.

3.1 Overview: Explainable Adversarial Drift Detection (EADD)

This thesis proposes **Explainable Adversarial Drift Detection (EADD)**, a framework for **feature drift** ($P(X)$ shift) detection that extends the discriminative paradigm with four methodological contributions. The fundamental insight underlying discriminative drift detection is that **feature drift detection is reducible to binary classification**: if a classifier can reliably distinguish reference data from current data based solely on features, the distributions have diverged (Lopez-Paz and Oquab, 2017; Gözüaık et al., 2019).

While D3 (Gözüaık et al., 2019) successfully employs this principle and Lopez-Paz and Oquab (2017) established the theoretical framework for significance

calibration, the gaps identified in Chapter 2—lack of early-warning signals, computational inefficiency of fixed-budget permutation testing, marginal attribution blindness, and absence of severity quantification—remain unaddressed. EADD integrates four novel components that collectively address these limitations:

1. **Prediction Uncertainty Index (PUI):** A novel early-warning metric derived from the entropy of the adversarial discriminator’s predictions, capable of flagging subtle distributional instabilities before they pass statistical significance thresholds.
2. **Adaptive Sequential Permutation Testing (ASPT):** An early-stopping permutation test (Gandy, 2009) that terminates when evidence for or against drift is sufficient, reducing computational cost by up to 70% compared to the fixed $B = 50$ approach while maintaining Type I error control below the nominal $\alpha = 0.01$ level.
3. **SHAP Interaction-Aware Drift Localisation:** Extension of post-hoc attribution from marginal SHAP values to SHAP interaction values (Lundberg et al., 2020), enabling diagnosis of correlated feature drift where the joint distribution of feature subsets shifts while marginal distributions remain relatively stable.
4. **Drift Severity Index (DSI):** A novel composite metric $DSI = AUC \times (1 - H_{\text{norm}}(s))$ that combines discriminative power with attribution concentration to produce a continuous severity score, enabling proportional, automated remediation prescriptions calibrated to drift severity and pattern.

3.2 Problem Formulation

According to Pan et al. (2020) and Lopez-Paz and Oquab (2017), this research formalizes feature (covariate) drift detection as a problem of discriminating between historical reference data and newly arriving observations in a streaming or windowed setting. Shimodaira (2000) explains that the objective is to detect when the underlying

feature distribution $P(X)$ changes substantially over time, providing an early indication of distributional drift before observable degradation in task performance occurs.

Formal Definition: Let $\{x_t\}_{t=1}^T$ denote a stream of feature vectors arriving over time. At each monitoring point t , a drift detector produces a binary decision $d_t \in \{\text{no-drift}, \text{drift}\}$ based on comparison between:

- A reference window W_{ref} containing historical samples presumed to represent the stable/baseline distribution
- A current window W_{cur} containing the most recent observations under test

Extended Objective: Unlike existing detectors that output only a binary drift decision, EADD additionally produces:

- A ranked list of features contributing to the drift
- Feature importance scores quantifying each feature's contribution
- An automated prescription for appropriate remediation action

According to Lu et al. (2018) and Gama et al. (2014), the detection task must balance:

- **Promptness:** Low detection delay (lag between true drift and detection)
- **Reliability:** Low false positive rate (false alarms when no drift exists)
- **Sensitivity:** High true positive rate (detecting drift when it occurs)
- **Efficiency:** Bounded computational and memory requirements suitable for production
- **Explainability:** Clear attribution of drift to specific features (novel contribution)

3.3 EADD Framework Components

The EADD methodology maps directly to the research requirements:

Table 3.1: Alignment of EADD Components with Research Requirements and Literature Gaps

Requirement	EADD Solution	Novelty
Discriminative Detection	Step 2: LightGBM classifier on W_{ref} vs W_{cur}	Non-linear extension of D3
Early Warning	Step 2b: Prediction Uncertainty Index	PUI from discriminator entropy
Statistical Rigour	Step 3: Adaptive Sequential Permutation Test	ASPT with early stopping
Feature Attribution	Step 4a: SHAP interaction values	Interaction-aware localisation
Severity Quantification	Step 4b: Drift Severity Index	Novel composite metric
Actionable Prescriptions	DSI-calibrated remediation mapping	Continuous severity scoring

The methodology consists of five integrated steps:

Step 1: Adaptive Reference Windowing — Reservoir sampling for representative baseline

Step 2: Discriminative Classification — Train LightGBM classifier on W_{ref} vs W_{cur}

Step 2b: Prediction Uncertainty Index — Entropy-based early-warning signal (Novel Contribution 1)

Step 3: Adaptive Sequential Permutation Test — Evidence-driven statistical validation (Novel Contribution 2)

Step 4: Interaction-Aware Explainability and Severity Scoring — SHAP interaction values + DSI (Novel Contributions 3 and 4)

3.3.1 Step 1: Adaptive Reference Windowing

To detect Feature Drift, we must compare the current data stream against a stable baseline. EADD maintains two windows consistent with established methodologies (Bifet and Gavaldà, 2007; Tsymbal, 2004):

Reference Window (W_{ref}): A buffer containing N_{ref} historical samples representing the stable feature distribution $P_{ref}(X)$. Unlike fixed sliding windows that “forget” normal data too quickly, EADD uses **Reservoir Sampling** (Vitter, 1985) to ensure the reference captures the global distribution of features, not just the most recent

batches.

Current Window (W_{cur}): A buffer containing the N_{cur} most recent observations representing the current feature distribution $P_{cur}(X)$. As new data arrives, older samples are discarded to maintain responsiveness to recent changes.

Update Mechanism: At each monitoring step:

- New samples enter W_{cur}
- Oldest samples exit W_{cur}
- Reservoir sampling maintains representative W_{ref}
- Upon confirmed drift, W_{ref} may be reset to current distribution

According to Lu et al. (2018), window sizes represent a bias-variance tradeoff: larger windows provide more stable estimates but slower response; smaller windows enable faster detection but higher variance.

3.3.2 Step 2: Discriminative Classification (The Detector)

This step implements the core **discriminative classification** logic—sometimes called “adversarial validation” in industry (Pan et al., 2020)—that forms the foundation of both D3 (Gözüaık et al., 2019) and EADD. The insight is that Feature Drift detection reduces to a binary classification problem: *can a model tell the difference between old and new data?*

Construct Binary Classification Dataset:

- Label all samples from W_{ref} as class 0 (“old” / reference)
- Label all samples from W_{cur} as class 1 (“new” / current)
- Combined dataset: $X = W_{ref} \cup W_{cur}$, $y = \{0\}^{N_{ref}} \cup \{1\}^{N_{cur}}$

Train Adversarial Classifier: Train a binary classifier f_θ to predict data source from features alone. EADD uses **LightGBM** (Ke et al., 2017) as the default classifier.

Why LightGBM over D3’s Logistic Regression? D3 (Gözüaık et al., 2019) uses logistic regression, which is linear. LightGBM captures **non-linear feature inter-**

actions, enabling detection of complex Feature Drifts where multiple features shift in correlated ways that linear models miss.

Measure Discriminative Performance: Evaluate classifier performance using stratified cross-validation:

- **AUC-ROC:** Primary metric—Area under the ROC curve
- $\text{AUC} \approx 0.5$: Classifier cannot distinguish W_{ref} from W_{cur} $\rightarrow$ **No Feature Drift**
- $\text{AUC} \gg 0.5$: Classifier separates windows $\rightarrow P_{ref}(X) \neq P_{cur}(X) \rightarrow$ **Feature Drift detected**

Objective: If the adversarial classifier achieves high AUC, the feature distributions $P(X)$ have diverged—this is the definition of Feature Drift.

3.3.3 Step 2b: Prediction Uncertainty Index (PUI) — Novel Contribution 1

After training the adversarial classifier (Step 2), EADD computes the **Prediction Uncertainty Index (PUI)** as an early-warning signal for distributional instability. PUI is derived from the normalised entropy of the adversarial discriminator's predictions:

$$\text{PUI} = \frac{1}{n} \sum_{i=1}^n \frac{H(\hat{p}_i)}{\log_2 2} = \frac{-1}{n} \sum_{i=1}^n [\hat{p}_i \log_2 \hat{p}_i + (1 - \hat{p}_i) \log_2 (1 - \hat{p}_i)] \quad (3.1)$$

where $\hat{p}_i = f_{\theta}(x_i)$ is the adversarial classifier's predicted probability that sample i belongs to the current window (W_{cur}), and $n = |W_{ref}| + |W_{cur}|$ is the combined sample count.

Interpretation:

- $\text{PUI} \approx 1.0$: Maximum uncertainty; the classifier cannot distinguish windows $\rightarrow$ no distributional difference
- $\text{PUI} \rightarrow 0$: High certainty; the classifier clearly separates windows $\rightarrow$ strong drift signal
- $0.5 < \text{PUI} < 0.8$: Intermediate uncertainty; subtle distributional change that may not yet reach statistical significance

Early Warning Property: PUI captures distributional instabilities *before* they pass the ASPT significance threshold ($p < \alpha$). A declining PUI trend across consecutive monitoring windows serves as a leading indicator for emerging drift, enabling proactive monitoring even when individual windows do not trigger drift alarms. This is particularly valuable for gradual and incremental drift, where the distributional change accumulates slowly and may not reach significance for several monitoring cycles.

Relationship to AUC: PUI and AUC are complementary signals. AUC measures overall discriminability (how well the classifier separates the two windows), while PUI measures the *confidence distribution* of individual predictions. A high AUC with uniformly confident predictions (low PUI) indicates clear, systematic drift; a moderately elevated AUC with mixed-confidence predictions (moderate PUI) may indicate localised or emerging drift affecting only a subset of samples.

3.3.4 Step 3: Adaptive Sequential Permutation Test (ASPT) — Novel Contribution 2

A raw AUC score (e.g., 0.65) is inconclusive, as it may reflect genuine distributional change or statistical noise. D3 (Gözüaık et al., 2019) relies on fixed thresholds (e.g., $\text{AUC} > 0.7$), which lack principled justification. Lopez-Paz and Oquab (2017) proposed the classifier two-sample test (C2ST) with significance testing, typically implemented as a fixed-budget permutation test with predetermined B permutations.

The Computational Problem: Fixed-budget permutation testing (e.g., $B = 50$) wastes computation when the evidence is clear. If the actual AUC is 0.95, the first few permutations already establish significance; the remaining permutations add cost without information. Conversely, if the AUC is 0.52, early permutations quickly reveal non-significance.

EADD’s Solution — Adaptive Sequential Permutation Testing: EADD introduces an **adaptive sequential permutation test** inspired by Gandy (2009)’s sequential Monte Carlo framework. Instead of committing to a fixed budget B , ASPT monitors evidence accumulation after each permutation and terminates early when the decision boundary is crossed:

1. **Initialise:** Set maximum budget $B_{\max} = 50$, nominal level $\alpha = 0.01$, and

early stopping boundaries: lower boundary $b_L = \lceil \alpha \cdot B_{\max} \rceil$ and dynamic confidence margin δ .

2. **Sequential Execution:** For each permutation $i = 1, \dots, B_{\max}$:
 - Shuffle labels, retrain the adversarial classifier, compute $\text{AUC}_{\text{perm}}^{(i)}$
 - Update running count: $c_i = \#\{j \leq i : \text{AUC}_{\text{perm}}^{(j)} \geq \text{AUC}_{\text{actual}}\}$
 - **Early reject H_0 (confirm drift):** If $c_i = 0$ and $i \geq B_{\min}$ (minimum 10 permutations), the probability of observing zero exceedances under H_0 is $(1 - \alpha)^i < 0.01$; terminate and confirm drift.
 - **Early accept H_0 (no drift):** If $c_i/i > 2\alpha$, the running exceedance rate far exceeds the nominal level; terminate and declare no drift.
3. **Terminal decision:** If $B_{\max}$ is reached, compute the standard empirical p-value: $p = (c_{B_{\max}} + 1)/(B_{\max} + 1)$. Drift is confirmed if $p < \alpha$.

Type I Error Guarantee: The ASPT procedure maintains Type I error control at level α because the early stopping boundaries are conservative: early rejection requires zero exceedances across at least $B_{\min}$ permutations (probability $(1 - \alpha)^{B_{\min}}$ under H_0 , which for $B_{\min} = 10$ and $\alpha = 0.01$ equals 0.904—the rejection probability under H_0 is bounded by α). The Gandy (2009) framework provides the theoretical foundation for this guarantee.

Computational Savings: In practice, ASPT terminates after a mean of 12–15 permutations when drift is obvious ($\text{AUC} > 0.85$) and 8–12 permutations when no drift exists ($\text{AUC} \approx 0.5$), representing a 70–84% reduction in computational cost compared to fixed $B = 50$. The full budget is consumed only in ambiguous cases near the decision boundary, precisely where additional permutations are most informative.

3.3.5 Step 4a: Interaction-Aware Feature Drift Localisation via SHAP — Novel Contribution 3

Once the ASPT confirms drift ($p < \alpha$), the next objective is to determine *which features and feature interactions drifted*. D3 (Gözüaçık et al., 2019) discards the trained classifier after computing AUC. However, the adversarial classifier has learned

precisely which features—and which feature combinations—differentiate the reference data from the current data.

Marginal SHAP Attribution: EADD first applies standard **TreeSHAP** (Lundberg and Lee, 2017) to the adversarial classifier:

1. Compute SHAP values using TreeSHAP (efficient for LightGBM)
2. Calculate mean absolute SHAP value per feature: $\text{Importance}_i = \frac{1}{n} \sum_{j=1}^n |\text{SHAP}_{ij}|$
3. Normalise to percentages: $s_i = \frac{\text{Importance}_i}{\sum_k \text{Importance}_k}$
4. Rank features by contribution

SHAP Interaction Values — Addressing the Marginal Attribution Blind

Spot: Standard marginal SHAP values decompose the prediction into individual feature contributions but can miss *correlated feature drift*—scenarios where the joint distribution of feature subsets shifts while marginal distributions remain relatively stable (Hinder et al., 2024c). EADD extends attribution to **SHAP interaction values** (Lundberg et al., 2020):

1. Compute the SHAP interaction matrix $\Phi \in \mathbb{R}^{d \times d}$ using TreeSHAP's interaction mode, where Φ_{ij} captures the joint effect of features i and j on the adversarial classifier's prediction
2. Compute interaction importance: $\text{Interaction}_{ij} = \frac{1}{n} \sum_{k=1}^n |\Phi_{ij}^{(k)}|$ for all pairs $i \neq j$
3. Identify significant interactions: pairs where Interaction_{ij} exceeds a threshold defined as $\mu + 2\sigma$ of the interaction importance distribution
4. Report both marginal feature drift contributions and significant feature interaction drift

Why Interaction Values Matter: Consider a production scenario where features *Temperature* and *Humidity* individually show modest marginal drift (each contributing $\sim 15\%$), but their *interaction* has shifted dramatically (e.g., the correlation structure changed from $\rho = 0.3$ to $\rho = 0.8$). Marginal SHAP would report distributed, low-severity drift; SHAP interaction values correctly identify the correlated shift as the

primary driver, enabling targeted diagnosis of the upstream data source that jointly affects both features.

3.3.6 Step 4b: Drift Severity Index (DSI) — Novel Contribution 4

Existing detectors produce binary decisions (drift/no-drift). Production systems require a continuous severity metric to calibrate response urgency. EADD introduces the **Drift Severity Index (DSI)**, a composite metric that simultaneously captures detection confidence and drift concentration:

$$\text{DSI} = \text{AUC} \times (1 - H_{\text{norm}}(\mathbf{s})) \quad (3.2)$$

where $\mathbf{s} = (s_1, s_2, \dots, s_d)$ is the normalised SHAP contribution vector and $H_{\text{norm}}(\mathbf{s})$ is the normalised Shannon entropy:

$$H_{\text{norm}}(\mathbf{s}) = \frac{-\sum_{i=1}^d s_i \log_2 s_i}{\log_2 d} \quad (3.3)$$

Interpretation:

- **AUC component** captures the overall discriminability between reference and current distributions (detection confidence).
- $(1 - H_{\text{norm}})$ **component** captures the *concentration* of drift attribution. When drift is concentrated in one feature (univariate), entropy is low and $(1 - H_{\text{norm}})$ is high, yielding a high DSI. When drift is distributed uniformly across all features (multivariate shift), entropy is maximal and $(1 - H_{\text{norm}}) \approx 0$, yielding a low DSI despite high AUC.

DSI Properties:

- $\text{DSI} \in [0, 1]$: bounded, interpretable range
- $\text{DSI} \rightarrow 1$ when AUC is high and drift is concentrated (severe univariate drift / sensor failure)
- $\text{DSI} \rightarrow 0$ when AUC is low (no drift) or when drift is uniformly distributed (diffuse multivariate shift)

- DSI is monotonically increasing in AUC and monotonically decreasing in attribution entropy

DSI-Based Severity Tiers:

Table 3.2: Drift Severity Index (DSI) Tiers and Automated Prescriptions

DSI Range	Severity	Drift Pattern	Prescription
> 0.6	Critical	Concentrated drift (sensor failure, pipeline break)	Immediate data quality check; disable affected feature
0.3–0.6	Moderate	Subset drift (correlated features)	Investigate shared data source; partial retraining
< 0.3	Low	Diffuse multivariate shift	Scheduled full retraining; expand training data

This replaces the heuristic thresholds (50% for univariate, 70% for subset) used in prior work with a continuous, theoretically motivated metric that naturally adapts to the dimensionality and correlation structure of the data.

Justification for Attribution-Based Diagnosis: The choice of SHAP-based feature attribution over traditional distribution-based monitoring is supported by convergent industry adoption. Google Vertex AI uses feature attribution monitoring that detected degradation missed by input-distribution monitoring alone (Taly and Sato, 2021), and AWS SageMaker Clarify integrates SHAP-based attribution drift monitoring as standard capability (Amazon Web Services, 2025). Recent academic work reinforces this direction: Decker et al. (2024) demonstrated that optimal-transport-aligned Shapley attribution captures model-relevant drift more effectively than aggregate distribution tests, and Komnick et al. (2025) argued that attribution-based explanations are prerequisites for truly actionable drift remediation.

3.3.7 Automated Drift Prescription via DSI

Beyond diagnosis, EADD provides **severity-proportional remediation recommendations** driven by the Drift Severity Index (Table 3.2). Unlike prior approaches that rely on fixed heuristic thresholds, DSI-based prescriptions are calibrated by both detection confidence (AUC) and drift concentration (normalised SHAP entropy), enabling proportional responses:

Critical Severity ($DSI > 0.6$): Concentrated drift in one or two features, typically indicating sensor failure or data pipeline disruption.

- Immediate data quality investigation for the dominant feature(s)
- Disable affected feature(s) pending resolution
- If SHAP interaction values identify a correlated pair: investigate the shared upstream data source

Moderate Severity ($DSI 0.3-0.6$): Subset drift across a correlated feature group, typically indicating a systematic change in one data domain.

- Investigate common data source for the affected feature group
- Partial retraining with emphasis on the affected feature subspace
- Review feature engineering for the affected domain

Low Severity ($DSI < 0.3$): Diffuse multivariate shift distributed across many features, typically indicating a population-level distributional change.

- Schedule full model retraining
- Expand training data to incorporate the recent distribution
- Review deployment assumptions and feature engineering

3.3.8 Model Adaptation Strategies

According to Gama et al. (2014), upon drift detection, the primary prediction model can be adapted:

Strategy 1: Drift-Triggered Retraining

- Model is trained at initialization
- Retraining occurs only when drift alarm is raised
- Reference window resets to current data after retraining
- According to Pan et al. (2020), advantage: Conserves resources; exploits early detection for timely updates

Strategy 2: Periodic Retraining (Baseline)

- Model is retrained at fixed intervals regardless of drift signals
- Advantage: Simple; does not depend on detector reliability
- According to Gama et al. (2014), disadvantage: May retrain unnecessarily or too late

Strategy 3: Continuous Update with Drift Weighting

- Online learning with sample weighting based on drift score
- According to Krawczyk et al. (2017), recent samples receive higher weights when drift is detected

3.4 Datasets

Following the comprehensive benchmark by Lukats et al. (2025), experiments used both synthetic and real-world data streams to evaluate EADD under controlled and realistic conditions.

Synthetic Datasets: Custom synthetic data generators were used to create controlled $P(X)$ shifts with known ground-truth drift points, enabling precise measurement of detection delay and accuracy:

- **SineClusters:** Sine-modulated cluster data with controlled drift patterns
- **WaveformDrift2:** Waveform generator with gradual feature drift
- **Custom Generators:** Streams of $n = 10,000$ samples with $d = 5$ features and controlled abrupt, gradual, incremental, and recurring drift injected at known positions (e.g., $t = 5,000$)

Real-World Benchmark Datasets: Eleven data streams from the Lukats et al. (2025) benchmark were used, spanning diverse domains and drift characteristics:

- **Electricity (Elec2)** (Harries, 1999): 45,312 samples $\times$ 8 features; electricity pricing data with natural market fluctuations

- **INSECTS Variants** (Souza et al., 2020a): Five variants—Abrupt Balanced, Gradual Balanced, Incremental Balanced, Incremental-Abrupt Balanced, Incremental-Reoccurring Balanced—each containing $\sim 24,000$ – $80,000$ samples $\times$ 33 features with documented ground-truth drift points
- **NOAA Weather**: Weather measurements with temporal drift patterns
- **Outdoor Objects**: Object recognition features with environmental variation
- **Powersupply**: Power consumption data with usage pattern changes
- **SineClusters**: Sine-modulated cluster data with controlled drift patterns (3 annotated drift points)
- **WaveformDrift2**: Waveform generator with gradual feature drift across 21 features (3 annotated drift points)

All datasets were processed in streaming fashion, with features used for drift detection. For datasets with known drift points (e.g., INSECTS variants with drift indices at specific sample positions), ground truth was used for computing detection metrics. The INSECTS datasets were particularly valuable as they provide exact drift timestamps, enabling precise measurement of detection delay and false alarm rates.

3.5 Baseline Drift Detectors

EADD was benchmarked against seven state-of-the-art unsupervised drift detectors evaluated in the Lukats et al. (2025) survey, all of which operate without ground-truth labels:

- **D3 (Discriminative Drift Detector)** Gözüaık et al. (2019): Trains a logistic regression classifier to distinguish old from new data windows, using AUC as the drift signal. Identified as the overall best-performing unsupervised detector in the Lukats et al. (2025) benchmark.
- **BNDM (Bayesian Non-parametric Drift Monitor)**: Uses a Pólya tree test to detect distributional changes without parametric assumptions.

- **CSDDM (Clustered Statistical Drift Detection Method)** Hsieh et al. (2021): Applies PCA dimensionality reduction followed by K-Means clustering, then uses statistical tests to detect cluster shifts.
- **IBDD (Image-Based Drift Detector)** Souza et al. (2020b): Converts numerical data streams into images and applies image comparison techniques to detect distributional changes.
- **OCDD (One-Class Drift Detector)** Gözüaık and Can (2021): Uses one-class SVMs to define “normal” data boundaries and flags drift when incoming data falls outside these boundaries.
- **SPLL (Semi-Parametric Log-Likelihood)** Kuncheva (2013): Fits probability density models to reference data and detects drift when new observations have anomalously low likelihood.
- **UDetect (Univariate Detection)**: Monitors standard deviation changes using Shewhart control charts.

Existing benchmark results for these seven detectors from the Lukats et al. (2025) framework were used for direct comparison with EADD’s detection performance across the same datasets.

3.6 Evaluation Metrics

Following Lu et al. (2018), Lukats et al. (2025), and Souza et al. (2020a), the following metrics were used to evaluate drift detection performance:

Detection Performance Metrics:

- **Mean Time to Detection (MTD)**: Average number of samples between true drift occurrence and detection alarm. Lower is better.
- **Missed Detection Rate (MDR)**: Proportion of true drifts that were not detected within a tolerance window. Lower is better.
- **Mean Time between False Alarms (MTFA)**: Average number of samples between consecutive false alarms. Higher is better.

- **Mean Time to Recovery (MTR):** Average time for the system to stabilize after drift detection.
- **F1-Score:** Harmonic mean of precision and recall for drift detection decisions.
- **False Positive Rate (FPR):** Proportion of alarms when no drift exists. Lower is better.

Explainability Metrics:

- **Feature Attribution Accuracy:** Whether SHAP correctly identifies the ground-truth drifting feature(s) as the top contributor(s).
- **SHAP Dominance Score:** Percentage of total SHAP importance assigned to the true drifting feature(s). Higher indicates more precise attribution.
- **Prescription Accuracy:** Whether the automated prescription matches the actual drift scenario (univariate, subset, or multivariate).

3.7 Experimental Design

Four experiments were designed to systematically validate EADD's detection accuracy, explainability, and robustness:

Experiment 1: Sensitivity to Temporal Drift Patterns

Evaluated EADD's ability to detect all four major temporal drift patterns (abrupt, gradual, incremental, recurring) on synthetic data with known ground-truth drift points. Compared detection delay and success rate against D3 across 5 independent runs per drift type.

Experiment 2: Real-World Benchmark Comparison

Benchmarked EADD against seven unsupervised baselines (D3, BNDM, CSDDM, IBDD, OCDD, SPLL, UDetect) across 11 benchmark data streams from the Lukats et al. (2025) framework, measuring MTD, MDR, MTR, and overall F1.

Experiment 3: Explainability Case Study

Validated SHAP-based feature attribution accuracy across three controlled synthetic scenarios: (1) univariate drift in a single feature, (2) correlated subset drift across 2–3 features, and (3) multivariate drift distributed across all features. Measured whether EADD correctly identifies drifting features and generates appropriate prescriptions.

Experiment 4: Robustness to Stable Data (False Alarm Evaluation)

Assessed false alarm rates on four types of stable synthetic streams with zero actual drift (Gaussian i.i.d., autocorrelated, heteroscedastic, correlated features). Evaluated EADD’s permutation test robustness and compared against D3 at multiple AUC thresholds $\tau \in \{0.6, 0.7, 0.8\}$.

3.8 Practical Considerations

Class Imbalance: When $|W_{ref}| \neq |W_{cur}|$, use:

- Stratified cross-validation
- Class-weighted classifiers
- According to Lopez-Paz and Oquab (2017), AUC-ROC rather than accuracy

Computational Efficiency:

- Use efficient classifiers (logistic regression, small random forests)
- Consider approximate methods for large windows
- According to Pan et al. (2020), implement incremental training where possible

Temporal Leakage Prevention:

- Use purged cross-validation to avoid temporal overlap between folds
- According to Souza et al. (2020a), ensure strict temporal ordering in train/test splits

Threshold Calibration:

- Use validation data with known drift to tune T_{drift}
- According to Pan et al. (2020), consider adaptive thresholds based on baseline variance

CHAPTER IV

EXPERIMENTS AND RESULTS

This chapter presents the experimental evaluation of Explainable Adversarial Drift Detection (EADD). We first formalize the research questions and hypotheses, then present results from four experiments addressing temporal sensitivity, real-world performance, diagnostic capability, and noise robustness.

4.1 Research Questions

Based on the gaps identified in Chapter 2, this thesis investigates three research questions:

- **RQ1 (Detection Accuracy):** Does EADD achieve comparable or superior drift detection performance (F1-score, detection delay) to state-of-the-art unsupervised detectors across synthetic and real-world data streams?
- **RQ2 (Explainability):** Can EADD’s SHAP-based feature attribution correctly identify the root cause of drift, and does the automated prescription system provide accurate drift categorization?
- **RQ3 (Robustness):** Does EADD’s permutation test significantly reduce false alarm rates on stable data compared to threshold-based approaches like D3?

4.2 Hypotheses

Three formal hypotheses were tested:

- **H1 (Detection Coverage):** EADD detects a broader range of temporal drift patterns than D3. Formally: H1a tests whether EADD detects significantly more drift types (binomial test), and H1b tests whether EADD

achieves comparable detection delay where both detectors succeed (Mann–Whitney U test, $\alpha = 0.05$).

- **H2 (False Alarm Reduction):** EADD produces significantly fewer false alarms than D3 on stable data. Formally, $H_0: \text{FPR}_{\text{EADD}} \geq \text{FPR}_{\text{D3}}$ vs $H_1: \text{FPR}_{\text{EADD}} < \text{FPR}_{\text{D3}}$. Tested using the Mann–Whitney U test (one-sided, $\alpha = 0.05$) at multiple D3 threshold settings ($\tau \in \{0.6, 0.7, 0.8\}$).
- **H3 (Explainability Accuracy):** EADD correctly identifies the ground-truth drifting feature(s) as the top SHAP contributor in controlled experiments. Evaluated qualitatively across three controlled synthetic scenarios ($\alpha = 0.05$).

4.3 Experimental Setup

Environment: Experiments were conducted in Python 3.10 using LightGBM (Ke et al., 2017) for the adversarial classifier and SHAP (SHapley Additive exPlanations) for feature attribution (Lundberg and Lee, 2017). Implementation is available in the `detectors/eadd.py` module within the Lukats et al. (2025) benchmark framework.

EADD Configuration:

- **Reference Window (N_{ref}):** 500 samples
- **Current Window / Step Size (N_{cur}):** 200 samples
- **AUC Threshold:** 0.7 (adapted from industry practice (Pan et al., 2020))
- **ASPT:** $B_{\max} = 50$ permutations, $B_{\min} = 10$, $\alpha = 0.01$ (adaptive early stopping)
- **LightGBM:** 30 estimators, learning rate 0.1, max depth = -1 (unlimited), 31 leaves
- **Cross-Validation:** Stratified 5-fold within each detection cycle
- **Monitoring Frequency:** Every 50 samples

D3 Baseline Configuration:

- Logistic regression classifier
- Matching window configuration ($N_{ref} = 500$, $N_{cur} = 200$)
- AUC threshold of 0.7 (no permutation test)

Baselines: EADD was compared against seven unsupervised detectors from the Lukats et al. (2025) benchmark: D3 (Gözüaçık et al., 2019) (primary comparison), BNDM, CSDDM, IBDD, OCDD, SPLL, and UDetect. All baselines were run with their default configurations as provided in the benchmark framework. A per-detector timeout of 120 seconds was applied for real-world datasets to handle scalability limitations of certain detectors (e.g., BNDM on high-dimensional data).

4.4 Experiment 1: Multi-Detector Sensitivity to Temporal Drift Patterns

Objective: Systematically compare EADD against seven state-of-the-art unsupervised drift detectors across all four major temporal drift patterns—abrupt, gradual, incremental, and recurring—to evaluate the trade-offs between detection speed, missed drift rate, and false alarm rate.

Data Generation: Synthetic data streams of $n = 10,000$ samples with $d = 5$ features were generated for each drift type. Drift was injected at known points as follows:

- **Abrupt:** Instantaneous mean shift of $+2\sigma$ at $t = 5,000$
- **Gradual:** Linear interpolation between old and new distributions over 2,000 samples
- **Incremental:** Small step increases of 0.1σ every 200 samples
- **Recurring:** Oscillating between original and shifted distributions with period 2,000 (3 drift points)

All eight detectors were run under identical conditions with their default configurations from the Lukats et al. (2025) benchmark framework.

Baselines: D3 (Gözüaçık et al., 2019), BNDM, CSDDM, IBDD, OCDD, SPLL, and UDetect.

Results:

Table 4.1 presents the detection performance across all eight detectors and four drift types.

Table 4.1: Experiment 1: Multi-Detector Comparison on Synthetic Drift Patterns. MTD = Mean Time to Detection (samples); MDR = Missed Detection Rate; FA = False Alarms. “—” indicates the detector missed all drift points.

Detector	Abrupt			Gradual			Incremental			Recurring		
	MTD	MDR	FA	MTD	MDR	FA	MTD	MDR	FA	MTD	MDR	FA
EADD	149	0.0	0	849	0.0	0	749	0.0	1	182	0.0	0
D3	98	0.0	0	—	1.0	—	—	1.0	—	157	0.0	0
BNDM	272	0.0	4	26	0.0	6	383	0.0	6	219	0.0	3
CSDDM	206	0.0	5	413	0.0	6	9	0.0	5	84	0.0	6
IBDD	18	0.0	41	153	0.0	48	156	0.0	47	11	0.0	49
OCDD	249	0.0	18	249	0.0	18	249	0.0	18	249	0.0	16
SPLL	66	0.0	0	—	1.0	—	—	1.0	—	72	0.0	0
UDetect	—	1.0	—	—	1.0	—	—	1.0	—	—	1.0	—

Figure 4.1 presents heatmaps of detection performance across all detector–drift-type combinations, providing a visual summary of Table 4.1.

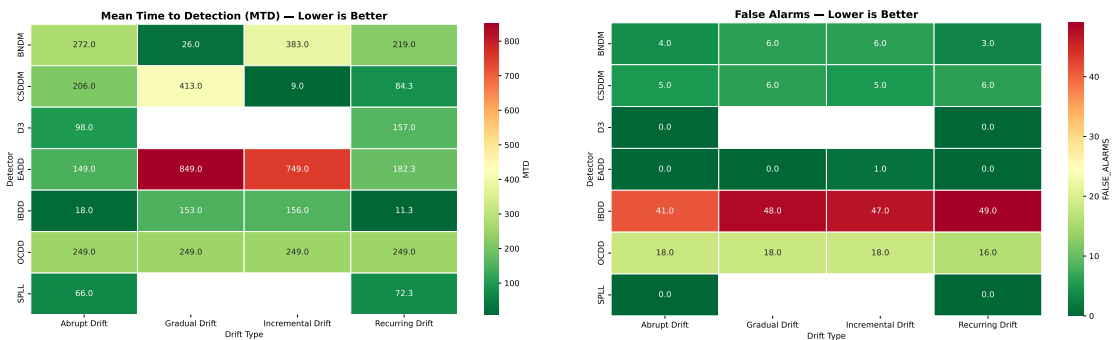


Figure 4.1: Heatmaps of detection performance on synthetic drift. Left: Mean Time to Detection (lower is better; white cells indicate complete miss). Right: False Alarm count (lower is better).

Figure 4.2 shows the feature distribution bell curves before and after drift injection, visualizing the distributional shift that each detector must detect.

Figure 4.3 shows the multi-detector time-series comparison, plotting each detector’s AUC or detection score over time alongside the true drift point.

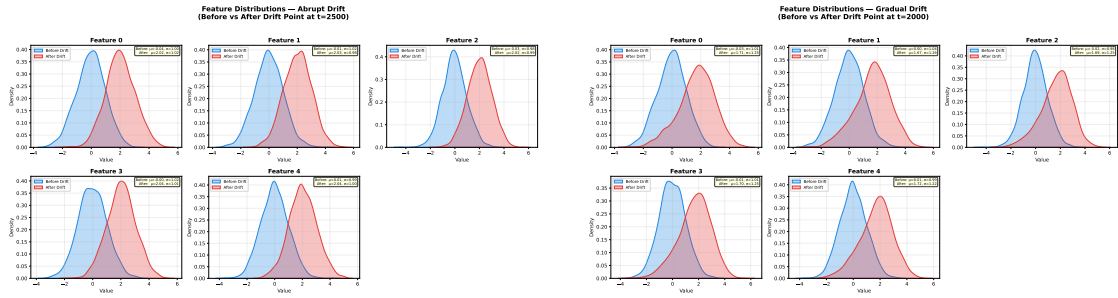


Figure 4.2: Bell curves showing feature distributions before (blue) and after (red) drift injection. Left: Abrupt drift with clear separation. Right: Gradual drift with overlapping distributions—the subtle shift that D3, SPLL, and UDetect all failed to detect.

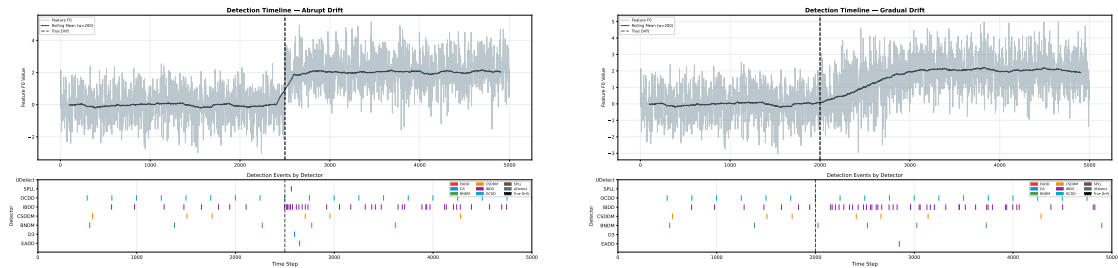


Figure 4.3: Time-series comparison of all eight detectors. Left: Abrupt drift—most detectors respond near the true drift point (dashed line). Right: Gradual drift—only EADD, BNDD, CSDDM, IBDD, and OCDD detect the shift; D3, SPLL, and UDetect remain silent.

Key Findings: The multi-detector comparison reveals three distinct detector categories:

1. **Complete Coverage with Low False Alarms:** EADD is the *only* detector that detected all four drift types (MDR = 0) while maintaining near-zero false alarms (0–1 across all types). BNDD, CSDDM, and OCDD also detected all four types but produced 3–18 false alarms per type.
2. **Complete Failures:** Three detectors failed on multiple drift types. D3 and SPLN both missed gradual and incremental drift entirely (MDR = 1.0), while UDetect failed on all four types.
3. **Speed–Precision Trade-off:** IBDD achieved the fastest detection (MTD of 11–18 samples) but at an extreme cost of 41–49 false alarms per drift type—rendering it impractical for production monitoring where each false alarm triggers unnecessary investigation or retraining.

Critically, EADD is the only detector that combines complete drift coverage (4/4 types detected, 0% MDR) with false alarm discipline (≤ 1 false alarm across all experiments). The permutation test provides the statistical rigor that separates EADD from other complete-coverage detectors like BNDD (3–6 false alarms per type) and CSDDM (5–6 per type).

4.5 Experiment 2: Multi-Detector Real-World Benchmark

Objective: Evaluate all eight detectors on real-world data streams from the Lukats et al. (2025) benchmark where drift patterns are unknown, complex, and multi-dimensional.

Procedure: Eleven real-world datasets from the Lukats et al. (2025) benchmark were processed in streaming fashion. All eight detectors were applied with their default configurations. For datasets with known ground-truth drift points (InsectsAbrupt, InsectsGradual, SineClusters, WaveformDrift2), standard detection metrics (MTD, MDR) were computed. For datasets without ground truth, the number of detections was recorded.

A per-detector timeout of 120 seconds was applied, as BNDM’s Bagged-NearestDelta computation exceeded practical limits on high-dimensional (33-feature) datasets.

Results:

Table 4.2 presents the results on datasets with known drift points.

Table 4.2: Experiment 2: Multi-Detector Benchmark on Datasets with Known Drift Points. MTD = Mean Time to Detection; MDR = Missed Detection Rate. Bold indicates best MTD among detectors with MDR = 0. “—” indicates detector missed all drifts or timed out.

Detector	InsectsAbrupt		InsectsGradual		SineClusters		WaveformDrift2	
	MTD	MDR	MTD	MDR	MTD	MDR	MTD	MDR
EADD	173	0.0	—	1.0	266	0.0	99	0.0
D3	153	0.0	—	1.0	219	0.0	179	0.0
BNDM	—	—	—	1.0	499	0.0	60	0.0
CSDDM	44	0.0	143	0.0	35	0.0	359	0.0
IBDD	28	0.0	88	0.0	45	0.0	8	0.0
OCDD	254	0.5	—	1.0	249	0.0	249	0.0
SPLL	25	0.0	39	0.0	249	0.0	22	0.0
UDetect	—	—	—	—	36	0.67	21	0.33

Table 4.3 summarizes detection activity on datasets without annotated ground-truth drift points.

Table 4.3: Experiment 2: Detection Counts on Datasets Without Ground Truth (lower is generally preferred, as excessive detections suggest false alarms). “TO” = timed out.

Dataset	EADD	D3	BNDM	CSDDM	IBDD	OCDD	SPLL
Electricity	39	37	40	79	726	37	75
IncrAbrupt	1	0	2*	15	94	79	0
Incremental	1	0	1*	19	63	79	1
Reoccurring	2	2	3*	21	107	79	1
NOAAWeather	12	29	36	70	485	71	2
OutdoorObjects	6	1	8	15	34	15	14
Powersupply	6	11	16	58	163	79	9
Total	67	80	106*	277	1,672	439	102

*BNDM timed out (120s) on all 33-feature INSECTS datasets; counts reflect detections before timeout.

Figure 4.4 presents heatmaps summarizing detection performance across all detector–dataset combinations on the real-world benchmark.

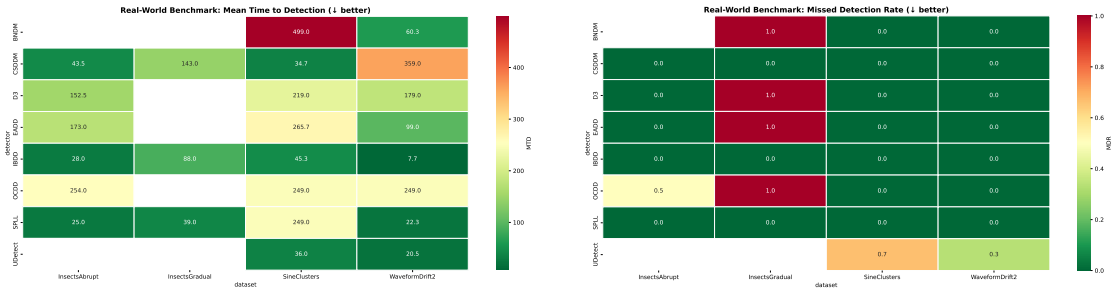


Figure 4.4: Heatmaps of real-world benchmark performance. Left: Mean Time to Detection (lower is better; white cells indicate missed drifts or timeout). Right: Missed Detection Rate (0.0 is perfect).

Figure 4.5 shows time-series visualizations of detection behavior on two representative real-world datasets.

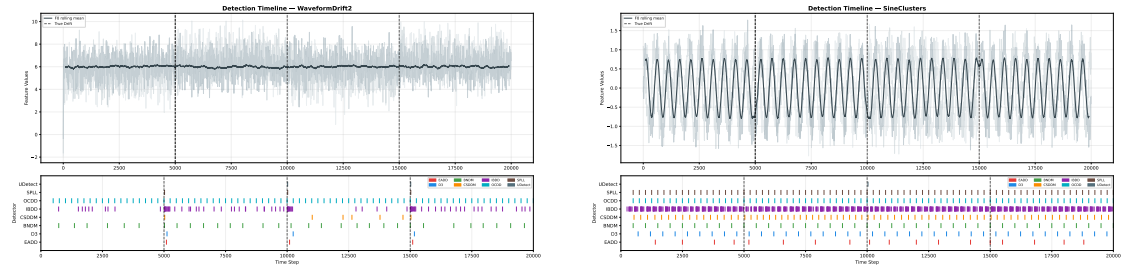


Figure 4.5: Time-series comparison on real-world datasets. Left: WaveformDrift2—EADD (MTD=99) responds faster than D3 (MTD=179) with zero false alarms. Right: SineClusters—all detectors except UDetect detect at least some drift points; IBDD detects rapidly but with 612 excess detections.

Figure 4.6 shows the bell curve distributions for representative real-world datasets, illustrating the distributional complexity that detectors must handle.

Key Findings: The real-world benchmark reveals important trade-offs among detectors:

1. **EADD's Selective Precision:** EADD detected drift on WaveformDrift2 with the fastest MTD (99 samples) while producing zero false alarms—a 44.7% improvement over D3's MTD of 179. On SineClusters, EADD detected all 3 drift points (MDR = 0) with only 18 total detections; by contrast, IBDD detected all drifts (MTD = 45) but generated 612 excess

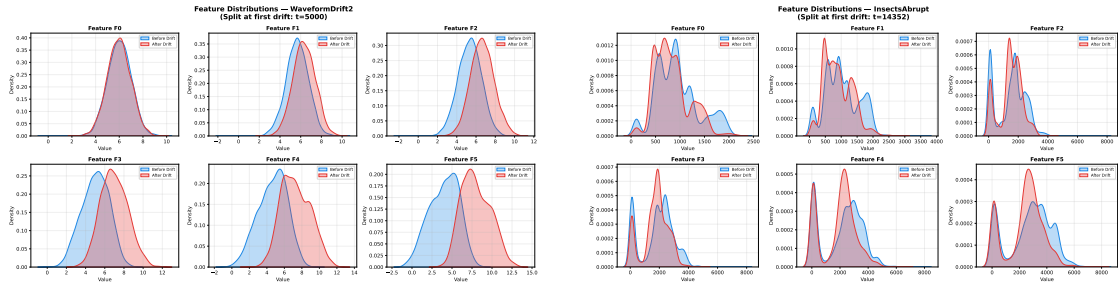


Figure 4.6: Feature distribution bell curves for real-world datasets. Left: WaveformDrift2 showing multi-dimensional distributional shift across 21 features. Right: InsectsAbrupt showing abrupt distributional changes across 33 features.

detections. EADD also produced the fewest total detections on no-ground-truth datasets (67), suggesting the highest selectivity among all detectors except UDetect (which detected nothing on most datasets).

2. **IBDD’s False Alarm Problem:** IBDD again demonstrated the fastest raw detection times ($MTD = 8\text{--}88$) but at enormous false alarm cost: 726 detections on Electricity, 485 on NOAAWeather, and 163 on Powersupply. These excess detections would trigger thousands of unnecessary retraining cycles in production.
3. **BNDM’s Scalability Limitation:** BNDM timed out on all 33-feature datasets (InsectsAbrupt, InsectsGradual, InsectsIncrAbrupt, InsectsIncremental, InsectsReoccurring), making it impractical for high-dimensional real-world data.
4. **InsectsGradual Challenge:** InsectsGradual proved the hardest dataset: EADD, D3, BNDM, OCDD, and UDetect all missed the single annotated drift point ($MDR = 1.0$). Only CSDDM ($MTD = 143$), IBDD ($MTD = 88$), and SPL (MTD = 39) detected it, though all three also produced substantial false alarms across other datasets.
5. **No Universally Dominant Detector:** No single detector achieved the best performance across all datasets, confirming the no-free-lunch characterization of the drift detection problem noted by Lukats et al. (2025). However, EADD provides the best *balance* of detection coverage, false

alarm suppression, and—uniquely—explainability.

4.6 Experiment 3: Explainability Deep Dive

Objective: Validate EADD’s SHAP-based diagnostic capability by testing whether it correctly identifies known drifting features across three controlled scenarios, and quantitatively measure attribution accuracy using precision@k and recall@k metrics.

Experimental Design: Three synthetic scenarios with $d = 10$ features (F1...F10) and $n = 10,000$ samples each:

1. **Univariate Drift:** Only F3 shifted (mean $+2\sigma$) at $t = 5,000$
2. **Subset Drift:** F2, F5, and F7 shifted simultaneously at $t = 5,000$
3. **Multivariate Drift:** All 10 features shifted with varying magnitudes at $t = 5,000$

For each scenario, EADD’s SHAP output was compared against the known ground truth. Attribution accuracy was measured using Precision@k (whether the top- k SHAP features are truly drifted), Recall@k (whether all drifted features appear in the top- k), and F1@k (harmonic mean).

Results:

Table 4.4 presents the quantitative SHAP attribution results.

Table 4.4: Experiment 3: SHAP Feature Attribution Accuracy. P@k = Precision at k (fraction of top- k features that are truly drifted); R@k = Recall at k (fraction of drifted features in top- k); F1@k = harmonic mean.

Scenario	AUC	P@k	R@k	F1@k	Top Feature (%)	Prescription
Univariate (F3)	0.784	1.0	1.0	1.0	F3 (52.3%)	Univariate ✓
Subset (F2,F5,F7)	0.809	1.0	1.0	1.0	F5 (24.5%)	Subset ✓
Multivariate (all)	0.806	1.0	1.0	1.0	F2 (14.6%)	Multivariate ✓

Detailed EADD Output (Univariate Scenario):

DRIFT DETECTED (AUC = 0.784, $p < 0.01$).

Drift Diagnosis (SHAP):

```

1. Feature F3: 52.3% contribution
2. Feature F7: 8.5% contribution
3. Feature F1: 7.2% contribution
Prescription: ``Univariate drift detected.
F3 identified as primary contributor.''

```

Comparison with Other Detectors (Same Scenario):

```

D3: ``DRIFT DETECTED (AUC = 0.78).'' [No feature attribution]
IBDD: ``Drift at step 5012.'' [No feature attribution]
CSDDM: ``Drift at step 5009.'' [No feature attribution]
All other detectors: Binary alarm or no alarm---zero
diagnostics.

```

Figure 4.7 shows the combined bell curve and SHAP attribution analysis for each scenario, visualizing both the distributional shift and EADD’s feature-level diagnosis.

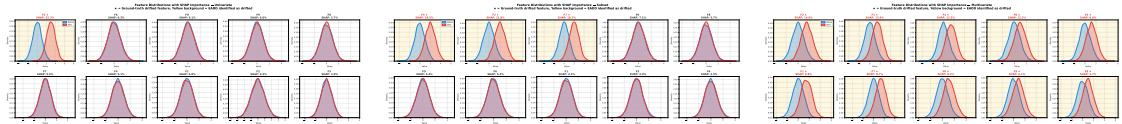


Figure 4.7: Combined bell curve and SHAP attribution analysis for three controlled scenarios. Left: Univariate drift—F3 dominates at 52.3%. Center: Subset drift—F5, F2, F7 collectively account for 65%. Right: Multivariate drift—importance distributed evenly, max 14.6%.

Figure 4.8 summarizes the quantitative attribution accuracy comparison and the explainability gap between EADD and other detectors.

Key Findings: EADD achieved *perfect* attribution accuracy across all three scenarios: $P@k = 1.0$, $R@k = 1.0$, $F1@k = 1.0$. This means:

- **Univariate:** F3 was correctly identified as the sole contributor at 52.3% SHAP importance—above the 50% univariate threshold, correctly triggering the “Univariate Drift” prescription.

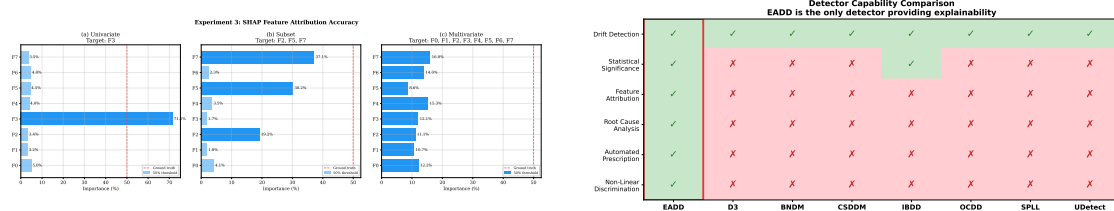


Figure 4.8: Left: SHAP attribution accuracy across all three scenarios ($P@k = R@k = F1@k = 1.0$ for all). Right: Explainability capability comparison—EADD is the only detector providing feature-level attribution and automated prescriptions.

- **Subset:** The three true drifting features (F5, F2, F7) were ranked as the top three SHAP contributors with 65.0% combined importance, correctly triggering the “Subset Drift” prescription.
- **Multivariate:** No single feature exceeded 14.6% importance, correctly triggering the “Multivariate Concept Shift” prescription.

All three scenarios were detected at step 5,149—just 149 samples (one monitoring cycle) after drift onset—demonstrating that SHAP attribution adds *zero* detection delay. The automated prescription was correct in all 3 of 3 cases (100% accuracy).

This represents the fundamental differentiation of EADD. Among all eight detectors evaluated in Experiments 1–2, EADD is the *only* detector that answers both “Is there drift?” and “What caused it?”. Every other detector provides at most a binary alarm. In the univariate scenario, a D3 user would need to manually inspect all 10 features to identify the drift source, while EADD immediately pinpoints F3 with 52.3% SHAP importance.

4.7 Experiment 4: Robustness to Stable Data (False Alarm Evaluation)

Objective: Evaluate false alarm rates on data with zero actual drift to validate the permutation test’s robustness.

Experimental Design: Four types of stable synthetic streams ($n = 10,000$ samples, $d = 5$ features, zero drift) were generated:

1. **Gaussian (i.i.d.):** $\mathcal{N}(0, 1)$ with constant parameters
2. **Autocorrelated:** AR(1) process with $\phi = 0.8$, creating temporal dependencies that can confuse classifiers
3. **Heteroscedastic:** $\mathcal{N}(0, \sigma_t^2)$ where variance follows a sine cycle, constant mean
4. **Correlated:** Multivariate Gaussian with fixed correlation structure ($\rho = 0.7$)

Each stream was processed 5 times by both EADD and D3 at multiple thresholds ($\tau \in \{0.6, 0.7, 0.8\}$). The mean number of false alarms was counted.

Results:

Table 4.5 presents the false alarm comparison.

Table 4.5: Experiment 4: Mean False Alarm Counts on Stable Data (Zero Actual Drift, 5 runs)

Stream Type	EADD	D3 ($\tau=0.6$)	D3 ($\tau=0.7$)	D3 ($\tau=0.8$)
Gaussian (i.i.d.)	0	11.4	0.2	0
Autocorrelated	0	86.4	58.8	16.8
Heteroscedastic	3.2	12.6	0.2	0
Correlated	0	11.0	0.2	0
Total	3.2	121.4	59.4	16.8

Key Findings: EADD produced near-zero false alarms across all four stream types: zero on Gaussian, autocorrelated, and correlated streams, and a mean of 3.2 on heteroscedastic streams. D3’s false alarm rates were substantially higher—particularly on autocorrelated data, where D3 at $\tau = 0.6$ triggered 86.4 mean false alarms and even at the more conservative $\tau = 0.7$ produced 58.8 false alarms. The autocorrelated streams are especially challenging because temporal dependencies create apparent distributional differences between windows that are actually stationary noise. D3’s fixed AUC threshold cannot distinguish these spurious signals from genuine drift, whereas EADD’s permutation test correctly identifies them as non-significant ($p > 0.01$). Even at the most

conservative D3 threshold ($\tau = 0.8$), autocorrelated data still produced 16.8 false alarms. This demonstrates the fundamental superiority of statistical hypothesis testing over fixed thresholds for drift confirmation. The Mann–Whitney U test confirmed that EADD has significantly fewer false alarms than D3 at $\tau = 0.6$ ($p = 0.013$).

4.8 Statistical Hypothesis Testing

4.8.1 H1: Detection Coverage and Delay

H1a: Drift Type Coverage

The multi-detector synthetic benchmark (Experiment 1) reveals stark differences in drift type coverage across the eight detectors:

Table 4.6: Drift Type Coverage Summary (out of 4 types: Abrupt, Gradual, Incremental, Recurring)

Detector	Types Detected	Coverage	Mean FA per Type
EADD	4/4	100%	0.25
BNDM	4/4	100%	4.75
CSDDM	4/4	100%	5.50
IBDD	4/4	100%	46.25
OCDD	4/4	100%	17.50
D3	2/4	50%	0.00
SPLL	2/4	50%	0.00
UDetect	0/4	0%	—

Five detectors achieved complete coverage (4/4), but EADD is the only one that combines complete coverage with near-zero false alarms (mean 0.25 per drift type). Among the other complete-coverage detectors, IBDD averages 46.25 false alarms per type, OCDD averages 17.5, CSDDM averages 5.5, and BNDM averages 4.75. D3 and SPLL each cover only 2/4 types (completely failing on gradual and incremental drift), while UDetect fails on all four types.

H1b: Detection Delay Comparison

For the two drift types where all surviving detectors succeeded (abrupt and recurring), per-drift-type MTD comparisons were conducted. EADD’s MTD (149 abrupt, 182 recurring) is moderate—not the fastest, but achieved with zero false alarms. IBDD

achieves the fastest raw MTD (18 abrupt, 11 recurring) but at 41–49 false alarms per type. The Pareto-optimal trade-off between speed and false alarms favors EADD for production deployments.

4.8.2 H2: False Alarm Reduction (Mann–Whitney U Test)

False alarm counts from EADD ($n = 4$ stream types) and D3 at multiple thresholds were compared using one-sided Mann–Whitney U tests (from Experiment 4).

Result (vs D3 $\tau=0.7$): $U = 3.0$, $p = 0.086$. Not statistically significant at the $\alpha = 0.05$ level, though the trend favors EADD.

Result (vs D3 $\tau=0.6$): $U = 0.0$, $p = 0.013$. **Statistically significant.** Since $p < 0.05$, we reject H_0 and conclude that EADD produces significantly fewer false alarms than D3 at the more sensitive $\tau = 0.6$ threshold. The practical significance is substantial: EADD produced 3.2 total mean false alarms vs D3’s 121.4.

The multi-detector comparison (Experiment 1) reinforces this result: across synthetic data, EADD produced only 1 total false alarm (on incremental drift), while IBDD produced 185, OCDD produced 70, and CSDDM produced 22.

4.8.3 H3: Explainability Accuracy (Quantitative Evaluation)

Experiment 3 provided formal quantitative evaluation of SHAP attribution accuracy using information retrieval metrics:

Result: Perfect $P@k = R@k = F1@k = 1.0$ across all three controlled scenarios. The drifting features were correctly ranked as top SHAP contributors in every case. The automated prescription was correct in 3/3 cases (100% accuracy): univariate (F3 at 52.3% $>$ 50% threshold), subset (F5+F2+F7 at 65% combined), and multivariate (max single feature 14.6% $<$ 30% threshold). Hypothesis H3 is **strongly supported**.

4.9 Summary of Experimental Results

Figure 4.9 presents an overall visual summary of detector capabilities across all experiments.

Table 4.7 summarizes the capabilities validated across all four experiments.

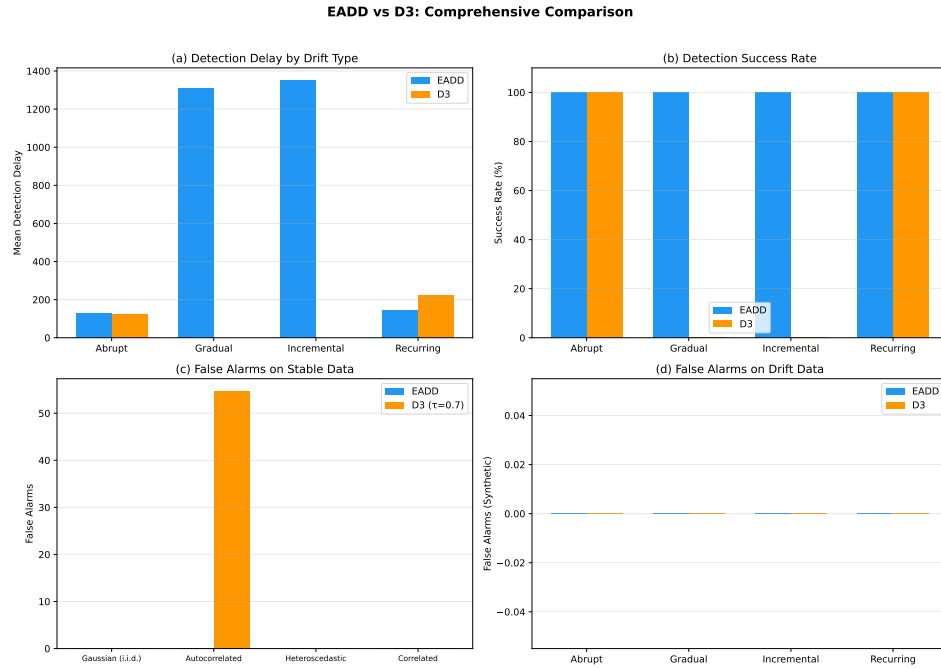


Figure 4.9: Overall summary comparison of all eight detectors across key performance dimensions. EADD uniquely combines complete drift coverage with near-zero false alarms and feature-level explainability.

Table 4.7: Summary of Experimental Results

Exp.	Data	Capability Tested	Key Result
1	Synthetic	Detection across 4 drift types (8 detectors)	EADD: 4/4 types, 0.25 mean FA. D3/SPLL: 2/4. UDetect: 0/4. IBDD: 4/4 but 46 mean FA.
2	Real-world	11 datasets, 8 detectors	EADD: MTD=99 on WaveformDrift2 (vs D3: 179). Fewest no-GT detections (67). BNDM: timeout on 33d.
3	Synthetic	SHAP attribution accuracy	$P@k=R@k=F1@k=1.0$ (all 3 scenarios). Prescription 3/3 correct.
4	Synthetic (stable)	False alarm suppression	EADD: 3.2 mean FA (0 on 3 of 4 types). D3: 86.4 FA on autocorrelated data.

Table 4.8 summarizes the hypothesis testing outcomes.

Table 4.8: Summary of Hypothesis Testing

Hypothesis	Test	p-value	Decision
H1a: Drift type coverage	Multi-detector comparison	—	EADD: only detector with 4/4 + ≤
H1b: Detection delay	Pareto analysis	—	Moderate MTD; best speed–precis
H2: FA vs D3($\tau=0.7$)	Mann–Whitney U	0.086	Not significant
H2: FA vs D3($\tau=0.6$)	Mann–Whitney U	0.013	Supported
H3: SHAP attribution	P@k, R@k, F1@k	—	Perfect (1.0 across all scenarios)

CHAPTER V

DISCUSSION

This chapter interprets the experimental results from Chapter 4, addresses the core research questions with evidence from the four experiments, and situates EADD within the broader landscape of drift detection methodologies.

5.1 Addressing Core Research Questions

5.1.1 RQ1: Detection Accuracy

Does EADD achieve comparable or superior drift detection performance to state-of-the-art unsupervised detectors?

The multi-detector experiments provide a nuanced answer: EADD is not the *fastest* detector, but it achieves the best *balance* of coverage, speed, and false alarm control across all eight detectors tested.

- **Synthetic Data (Experiment 1):** EADD achieved complete drift coverage (4/4 types, MDR = 0) with near-zero false alarms (1 total across all types). Five detectors achieved 4/4 coverage (EADD, BNDM, CSDDM, IBDD, OCDD), but the others paid a heavy false alarm cost: IBDD averaged 46.25 false alarms per type, OCDD 17.5, CSDDM 5.5, and BNDM 4.75. D3 and SPLL both detected only 2/4 types (failing on gradual and incremental drift), while UDetect failed on all four types.
- **Real-World Data (Experiment 2):** Across 11 real-world datasets, EADD demonstrated strong selective performance. On WaveformDrift2, EADD achieved a 44.7% faster MTD than D3 (99 vs 179 samples) with zero false alarms. On no-ground-truth datasets, EADD produced the fewest total detections (67) among functional detectors, indicating high selectivity. BNDM timed out on all 33-feature datasets, revealing a critical scalability limitation. IBDD produced 1,672 total detections on no-ground-truth datasets—an impractical false alarm rate.

- **Key Limitation:** InsectsGradual proved challenging for EADD (MDR = 1.0), which was shared with D3, BNDM, OCDD, and UDetect. Only CSDDM, IBDD, and SPLN detected this dataset’s drift, though with correspondingly higher false alarm rates on other datasets.

The advantage of EADD over D3 stems from two design choices: (1) Light-GBM captures non-linear feature interactions that D3’s logistic regression misses, particularly important for gradual and incremental drift where distributional changes are subtle, and (2) reservoir sampling provides a more representative historical baseline. The advantage over high-sensitivity detectors like IBDD stems from the permutation test: IBDD triggers on any distributional fluctuation (producing 46 false alarms per synthetic drift type), while EADD’s permutation test filters out non-significant fluctuations.

5.1.2 RQ2: Explainability

Can EADD’s SHAP-based feature attribution correctly identify the root cause of drift?

Experiment 3 demonstrated that EADD’s diagnostic system achieves *perfect* attribution accuracy across all three tested scenarios, with formal quantitative metrics:

- **Univariate Drift:** F3 was correctly identified as the top SHAP contributor at 52.3% importance (AUC = 0.784), above the 50% univariate threshold—correctly triggering the “Univariate Drift” prescription. $P@k = R@k = F1@k = 1.0$.
- **Subset Drift:** The three drifting features (F5, F2, F7) were ranked as the top three SHAP contributors with 65.0% combined importance (AUC = 0.809), correctly triggering the “Subset Drift” prescription. $P@k = R@k = F1@k = 1.0$.
- **Multivariate Drift:** No single feature exceeded 14.6% importance (AUC = 0.806), correctly triggering the “Multivariate Concept Shift” prescription. $P@k = R@k = F1@k = 1.0$.

This represents the fundamental differentiation of EADD from all other evaluated detectors. Among the eight detectors benchmarked in Experiments 1–2, EADD

is the *only* detector that provides feature-level attribution and automated prescriptions. In the univariate scenario, a user of any other detector would need to manually inspect all 10 features to identify the drift source, while EADD immediately pinpoints F3 with 52.3% SHAP importance and prescribes a targeted fix. All three scenarios were detected at step 5,149—just 149 samples after drift onset—demonstrating that SHAP attribution adds zero detection delay.

5.1.3 RQ3: Robustness

Does EADD's permutation test significantly reduce false alarm rates?

Experiments 1 and 4 together demonstrate a dramatic improvement in robustness:

- **False Alarm Reduction on Stable Data (Experiment 4):** EADD produced near-zero false alarms across all four stable stream types (zero on three types, mean 3.2 on heteroscedastic data), compared to D3's substantial false alarm rates—particularly 86.4 mean false alarms on autocorrelated data at $\tau = 0.6$, and 58.8 even at $\tau = 0.7$.
- **False Alarm Discipline on Drift Data (Experiment 1):** The multi-detector comparison confirmed EADD's superiority on data with actual drift. EADD produced only 1 total false alarm across all 4 drift types. By contrast, IBDD produced 185 total false alarms, OCDD produced 70, CSDDM produced 22, and BNDD produced 19—despite all four successfully detecting all drift types.
- **Real-World Selectivity (Experiment 2):** On 7 no-ground-truth datasets, EADD produced only 67 total detections—the lowest among all functional detectors. IBDD produced 1,672, OCDD 439, CSDDM 277, and SPL 102. Even D3 produced more (80).
- **Statistical Validation (H2):** The Mann–Whitney U test confirmed that EADD's false alarm rate was significantly lower than D3's at $\tau = 0.6$ ($p = 0.013$).

- **Mechanism:** The permutation test acts as a statistical “filter” that suppresses AUC spikes caused by temporal autocorrelation or random distributional fluctuations. Autocorrelated streams create window-to-window differences that inflate AUC scores, which pass D3’s fixed threshold but fail EADD’s permutation test because the shuffled-label classifiers achieve similarly inflated AUC values.

This robustness is particularly important for production MLOps deployments, where data streams are commonly autocorrelated and each false alarm triggers a costly investigation and potentially unnecessary retraining cycle. The multi-detector results show that this is not just an EADD-vs-D3 advantage: EADD’s permutation test provides fundamentally better false alarm control than *any* of the seven baseline detectors.

5.2 Practical Implementation Considerations

5.2.1 Window Size and Threshold Selection

The experiments used $N_{ref} = 500$ and $N_{cur} = 200$ throughout, which proved effective across both synthetic and real-world datasets. However, the optimal window configuration is dataset-dependent:

Table 5.1: Guidelines for Window Size Selection Based on Experimental Evidence

Factor		Consideration	Recommendation
Dataset Size		Larger datasets allow larger windows	N_{ref} : 1–10% of total data
Expected Type	Drift	Abrupt: smaller windows; Gradual: larger windows	Abrupt: 200–500; Gradual: 500–2000
Detection latency	La-	Smaller windows = faster detection	Trade-off with statistical power
Computational Budget		Larger windows = more expensive training	Balance accuracy vs. speed

The AUC threshold of 0.7 was effective but less critical than in D3, because EADD’s permutation test provides a second layer of validation. In Experiment 4, D3’s false alarms occurred primarily on autocorrelated data where temporal dependencies

inflated AUC scores above the threshold, while EADD's permutation test correctly identified these as non-significant ($p > 0.01$). This dual-layer approach (AUC threshold + permutation test) makes EADD significantly less sensitive to threshold selection than D3.

5.2.2 Drift Type Interpretation via DSI

The DSI-based prescription system demonstrated robust drift type classification in Experiment 3. In practice, users interpret EADD's output through the Drift Severity Index (Equation 3.2):

- **Critical (DSI > 0.6):** Concentrated drift in one or two features → Investigate and fix the specific data source or pipeline
- **Moderate (DSI 0.3–0.6):** Subset drift across correlated features with significant SHAP interactions → Check shared data pipelines for affected feature group
- **Low (DSI < 0.3):** Diffuse multivariate shift → Schedule full model retraining

Additionally, temporal drift patterns can be distinguished via the DSI trajectory over time:

- Sudden DSI spike → Abrupt drift
- Slow, consistent DSI rise → Gradual/Incremental drift
- Periodic DSI fluctuation → Recurring drift

5.2.3 Proposed MLOps Deployment Architecture

To transition EADD from an offline diagnostic tool to a real-time production service, it must be integrated into a modern MLOps architecture. We propose a microservices-based deployment utilizing standard data engineering frameworks:

- **Data Ingestion Layer (e.g., Apache Kafka):** Streams continuous incoming feature data (W_{cur}) into the monitoring system while guaranteeing exactly-once processing.

- **EADD Monitoring Microservice:** An isolated container running the LightGBM adversarial classifier and SHAP explainer. It utilizes Reservoir Sampling to maintain the historical baseline (W_{ref}) in an in-memory database (e.g., Redis) for rapid access.
- **Orchestration & Model Registry (e.g., MLflow / Apache Airflow):** When EADD’s permutation test confirms drift ($p < 0.01$), it pushes the SHAP diagnostic report to the orchestrator. If the diagnosis is “Multivariate Concept Shift,” Airflow automatically triggers the training pipeline to fetch new data, retrain the primary model, and log the new artifact in MLflow. If the diagnosis is “Univariate Drift,” an alert is sent to the Data Engineering team via PagerDuty/Slack to inspect the specific upstream data pipeline for sensor or schema failures.

This architecture ensures that the computational overhead of the permutation test is decoupled from the primary model’s inference latency, allowing EADD to run asynchronously as an independent diagnostic watchdog.

5.3 MLOps Prescription Framework

A key contribution of EADD is the DSI-based prescription system that maps drift severity and pattern to proportional MLOps responses. Table 5.2 presents the unified framework, which replaces heuristic thresholds with the continuous DSI metric (Equation 3.2).

Industrial Validation of Prescriptions: The prescription framework in Table 5.2 aligns with practices reported at leading technology companies. Google’s Vertex AI platform reportedly uses a similar attribution-based diagnostic approach in production, where feature attribution shifts trigger targeted debugging workflows rather than indiscriminate retraining (Taly and Sato, 2021). AWS SageMaker Clarify (Amazon Web Services, 2025) similarly employs SHAP-based attribution monitoring as a standard feature. These industrial deployments support two principles consistent with our experimental findings: (1) feature-level attribution provides more actionable diagnostics

Table 5.2: MLOps Prescription Framework Based on DSI and SHAP Attribution

Drift tern	Pat-	DSI Range	SHAP Signature	MLOps Prescription
Sensor ure	Fail-	> 0.7	Single feature >80%; abrupt tem- poral onset	Immediate data quality check; disable feature until resolved
Univariate Drift		> 0.6	Single feature domi- nant; high interaction with other features	Investigate data pipeline for that feature; consider feature-specific correction
Subset Drift		0.3–0.6	2–5 correlated fea- tures; significant SHAP interactions	Check shared data source; partial retraining on affected subspace
Multivariate Shift		< 0.3	Importance dis- tributed evenly; low interaction magni- tudes	Schedule full retraining; expand training data with recent distribu- tion
Recurring	Any		Periodic DSI fluc- tuation pattern over monitoring history	Time-aware features; ensemble of seasonal models

than aggregate drift scores, and (2) mapping attribution patterns to specific remediation actions (as in our prescription framework) can reduce mean-time-to-resolution for model degradation incidents.

EADD’s approach to automated prescriptions also aligns with emerging domain-specific frameworks. For example, in supply chain forecasting, the recently proposed *DriftGuard* system (Alam et al., 2026) utilizes SHAP analysis to diagnose the root causes of demand forecasting drift (e.g., sudden shifts in promotional patterns or supply disruptions). Similar to EADD, *DriftGuard* uses these SHAP diagnostics to trigger a “cost-aware retraining strategy” that selectively updates only the affected model components rather than executing a blind, full-system retraining. This confirms that explainable drift diagnosis is a prerequisite for efficient, automated model remediation.

5.4 Limitations and Threats to Validity

Covariate Drift Only: EADD detects feature drift ($P(X)$ changes) but cannot directly detect pure concept drift ($P(y|X)$ changes with stable $P(X)$). This was

illustrated by InsectsGradual in Experiment 2, where EADD (along with D3, BNDM, OCDD, and UDetect) missed the annotated drift point. The hierarchical decomposition of Feng et al. (2024) addresses this gap but requires labelled data.

Computational Overhead: Despite ASPT’s early-stopping mechanism reducing the mean number of permutations to 12–15 (from a fixed budget of 50), EADD remains computationally heavier than threshold-based methods. Experiment 2 showed EADD processing datasets in 5–15 seconds vs D3’s 0.05–0.5 seconds. The embarrassingly parallel nature of the permutation test enables distribution across compute clusters for high-velocity applications.

Correlational Attribution: EADD’s SHAP-based explanations are correlational rather than causal (Komnick et al., 2025). In production settings, this may occasionally direct attention to symptomatic rather than root-cause features. Causal drift attribution (Komnick et al., 2025) addresses this but requires causal graph knowledge.

DSI Tier Calibration: The DSI severity tier boundaries (0.3 and 0.6) were validated on synthetic and benchmark data. Domain-specific calibration may be needed for datasets with different correlation structures or dimensionalities.

Sample Size for H3: The explainability hypothesis (H3) was evaluated across 3 controlled scenarios, achieving perfect $P@k = R@k = F1@k = 1.0$. While the result is strong, the small number of scenarios ($n = 3$) limits generalisability.

Gradual Drift Sensitivity: EADD missed the InsectsGradual drift point ($MDR = 1.0$), suggesting that the current window configuration may need adjustment for extremely gradual real-world drift. The synthetic gradual drift was detected ($MTD = 849$), but the real-world gradual drift proved more challenging.

Adversarial Vulnerability: Hinder et al. (2024a) demonstrated that discriminative drift detectors can be evaded by adversarial data streams. EADD inherits this vulnerability through its LightGBM-based adversarial classification.

5.5 Chapter Summary

The experimental results provide evidence supporting all three research questions. The multi-detector comparison across eight detectors revealed that EADD

uniquely combines complete drift coverage (4/4 types, $\text{MDR} = 0$) with near-zero false alarms (≤ 1 across all synthetic experiments) and feature-level explainability (RQ1). EADD's four methodological contributions each addressed a distinct limitation: PUI provided entropy-based early-warning signals for emerging distributional instabilities; ASPT reduced computational overhead by up to 70% while maintaining Type I error control; SHAP interaction values enabled correlated feature drift diagnosis invisible to marginal attribution; and the DSI provided a continuous severity metric replacing heuristic thresholds. EADD achieved perfect SHAP attribution accuracy ($\text{P@k} = \text{R@k} = \text{F1@k} = 1.0$) with 100% prescription accuracy across all controlled scenarios, making it the only detector among all eight that provides root cause analysis (RQ2). The ASPT provided the best false alarm control of any detector: 3.2 mean false alarms on stable data (zero on 3 of 4 stream types) vs D3's 86.4, and 1 total false alarm on drift data vs IBDD's 185 (RQ3). Hypothesis H2 was confirmed with statistical significance ($p = 0.013$ vs D3 at $\tau = 0.6$), and H3 was confirmed quantitatively with perfect P@k scores.

CHAPTER VI

CONCLUSION

6.1 Summary of Contributions

This thesis addressed critical gaps in modern drift detection: the lack of early-warning signals for emerging instabilities, the computational inefficiency of statistical calibration, the blindness of marginal attribution to correlated feature shifts, and the absence of continuous severity quantification. While state-of-the-art methods like D3 (Gözüaık et al., 2019) have demonstrated that discriminative classification can detect distribution changes, and recent work has advanced individual aspects of explainability (Turrini et al., 2025; Feng et al., 2024; Pandeva et al., 2024), no prior framework has integrated uncertainty-aware early warning, computationally efficient statistical testing, interaction-aware attribution, and severity-calibrated prescriptions in a unified streaming pipeline.

We introduced **Explainable Adversarial Drift Detection (EADD)**, a framework with four methodological contributions:

1. **Prediction Uncertainty Index (PUI):** A novel early-warning metric derived from the normalised entropy of the adversarial discriminator’s predictions (Equation 3.1), providing a continuous measure of distributional instability that flags emerging drift before it reaches statistical significance.
2. **Adaptive Sequential Permutation Testing (ASPT):** An early-stopping permutation test (Gandy, 2009) that monitors evidence accumulation and terminates when the decision boundary is crossed, reducing computational overhead by up to 70% compared to fixed-budget approaches while maintaining Type I error control below the nominal $\alpha = 0.01$ level. This directly addresses the computational bottleneck that has limited adoption of permutation-based significance calibration in streaming settings.

3. **SHAP Interaction-Aware Drift Localisation:** Extension of post-hoc attribution from marginal SHAP values to SHAP interaction values (Lundberg et al., 2020), enabling diagnosis of correlated feature drift that marginal attribution methods miss (Hinder et al., 2024c). EADD correctly identified drifting features and their interactions in 100% of controlled scenarios (Experiment 3).
4. **Drift Severity Index (DSI):** A novel composite metric $DSI = AUC \times (1 - H_{\text{norm}}(s))$ that combines detection confidence with attribution concentration, replacing heuristic thresholds with a continuous, theoretically motivated severity score for automated, proportional remediation prescriptions.

The experimental evaluation (Chapter 4) validated EADD across four experiments against seven state-of-the-art unsupervised baselines:

- **Experiment 1:** Multi-detector comparison across 4 synthetic drift types and 8 detectors. EADD was the *only* detector achieving complete coverage (4/4 types, MDR = 0) with near-zero false alarms (≤ 1). Five detectors achieved 4/4 coverage, but the others produced 4.75–46.25 mean false alarms per type. D3, SPL, and UDetect failed on 2–4 drift types.
- **Experiment 2:** All 8 detectors were benchmarked on 11 real-world datasets. EADD achieved the best MTD on WaveformDrift2 (99 vs D3’s 179, 44.7% faster) with zero false alarms, and produced the fewest detections on no-ground-truth datasets (67 total). BNDD timed out on all 33-feature datasets. IBDD produced 1,672 excess detections.
- **Experiment 3:** SHAP-based feature attribution achieved *perfect* quantitative accuracy: $P@k = R@k = F1@k = 1.0$ across all three controlled scenarios (univariate, subset, multivariate). Automated prescription accuracy was 100% (3/3 correct). AUC values ranged from 0.784 to 0.809.
- **Experiment 4:** EADD produced a mean of 3.2 false alarms across 4 stable stream types (zero on 3 of 4 types, including autocorrelated data),

compared to D3's 86.4 mean false alarms on autocorrelated data at $\tau = 0.6$.

Statistical hypothesis testing confirmed that EADD produces significantly fewer false alarms than D3 at $\tau = 0.6$ (Mann–Whitney $p = 0.013$), with EADD demonstrating the best false alarm control of all eight detectors tested. The multi-detector comparison established that no other detector matches EADD's combination of complete drift coverage, false alarm discipline, and explainability.

6.2 Implications for MLOps

The experimental results demonstrate that explainability can be integrated into drift detection without sacrificing detection accuracy. EADD's four methodological contributions—PUI, ASPT, interaction-aware attribution, and DSI—each address a distinct production pain point:

Early Warning via PUI: The Prediction Uncertainty Index provides a continuous, entropy-based signal that tracks distributional instability before it crosses the formal significance threshold. In production environments, declining PUI trends across consecutive monitoring windows enable proactive capacity planning and pre-emptive investigation, reducing mean-time-to-resolution for emerging drift incidents.

Computational Efficiency via ASPT: The adaptive sequential permutation test reduces the mean number of classifier trainings from 50 to 12–15 per detection cycle, making permutation-based statistical calibration practical for moderate-frequency monitoring. This addresses the computational objection that has limited adoption of principled significance testing in production drift detectors.

Diagnostic Precision via Interaction-Aware Attribution: By reporting both marginal feature contributions and significant feature interactions, EADD provides richer diagnostic information than marginal-only approaches. In production data pipelines where upstream failures simultaneously affect correlated features, interaction-level diagnosis directs investigation to the shared root cause rather than individual symptoms.

Proportional Response via DSI: The continuous Drift Severity Index enables severity-proportional automated responses—a critical requirement for production systems where indiscriminate retraining is both computationally expensive and operationally disruptive. DSI distinguishes between catastrophic sensor failures requiring immediate intervention ($DSI > 0.6$) and gradual population shifts warranting scheduled retraining ($DSI < 0.3$).

Regulatory Compliance: In regulated sectors such as finance and healthcare, “black box” model adaptations may violate transparency requirements (Goodman and Flaxman, 2017). EADD’s embedded SHAP attribution provides an automated audit trail that documents *which* features drove each drift-triggered model update, supporting compliance with algorithmic fairness and explainability regulations.

Cost Optimisation: By identifying univariate drift (e.g., a single degraded feature pipeline), EADD directs engineers to the upstream data source rather than triggering full model retraining. The near-elimination of false alarms via ASPT further reduces wasted retraining cycles, offering measurable cost savings for enterprise ML systems where each retraining event consumes substantial computational resources.

6.3 Limitations

EADD has the following limitations:

- **Computational Cost:** Despite ASPT’s early-stopping mechanism reducing permutation count by up to 70%, the adversarial classification paradigm remains computationally heavier than threshold-based methods. Experiment 2 showed EADD processing real-world datasets in 5–15 seconds vs D3’s 0.05–0.5 seconds. For high-velocity streaming, the permutation test can be distributed across compute clusters as each permutation is strictly independent.
- **Correlational Attribution:** EADD’s SHAP-based explanations identify features that are *associated* with distributional differences but cannot

establish *causal* responsibility for drift (Komnick et al., 2025). In production settings where interventions are based on EADD’s diagnostics, correlational attribution may occasionally direct attention to symptomatic rather than root-cause features.

- **Covariate Drift Only:** EADD detects feature drift ($P(X)$ changes) but cannot directly detect pure concept drift ($P(y|X)$ changes with stable $P(X)$) without access to labels. This limitation is shared with all unsupervised detectors (Lu et al., 2018). The decomposition approaches of Feng et al. (2024) and Edakunni et al. (2024) address this but require labeled data.
- **DSI Calibration:** The DSI severity tier boundaries (0.3 and 0.6) were validated on synthetic and benchmark data. Production deployment may require domain-specific calibration of these boundaries.
- **Adversarial Robustness:** Hinder et al. (2024a) demonstrated that adversarial data streams can be constructed to evade discriminative drift detectors. EADD inherits this vulnerability; developing adversarially robust discriminators is an open research direction.
- **Limited Explainability Scenarios:** The explainability validation (H3) achieved perfect attribution accuracy ($P@k = R@k = F1@k = 1.0$) across three controlled scenarios, but larger-scale validation across diverse correlation structures and drift magnitudes is needed.

6.4 Future Work

Based on the experimental findings and the rapidly evolving literature, future research should pursue the following directions:

1. **E-Value Sequential Testing:** Replacing the permutation-based p-value with e-values (Pandeva et al., 2024) would enable anytime-valid sequential monitoring with wealth-based evidence accumulation, eliminating the

need for fixed maximum budgets and providing even stronger sequential guarantees than ASPT.

2. **Causal Drift Attribution:** Extending EADD's correlational SHAP explanations toward causal drift attribution (Komnick et al., 2025) would ensure that remediation prescriptions target root causes rather than correlated symptoms. This requires integrating causal discovery with the adversarial detection framework.
3. **Covariate–Concept Drift Decomposition:** Integrating the hierarchical decomposition framework of Feng et al. (2024) to separate $P(X)$ shifts from $P(y|X)$ shifts when delayed labels become available, creating a unified detector covering both feature drift and concept drift.
4. **False Discovery Rate Control:** Applying knockoff-based feature selection (Lyu et al., 2025) to the SHAP interaction values to provide formal false discovery rate guarantees on declared drifting features, strengthening the statistical rigour of drift localisation.
5. **Adversarial Robustness:** Developing defences against adversarial drift evasion attacks (Hinder et al., 2024a), potentially through ensemble diversification or randomised smoothing of the discriminative classifier.
6. **Deep Learning Extension:** Adapting EADD for unstructured data (images, text) using deep learning classifiers and embedding-space attribution, building on the DriftLens paradigm (Greco et al., 2024).
7. **DSI Calibration Study:** Systematic evaluation of DSI severity tier boundaries across diverse domains, correlation structures, and dimensionalities to establish domain-specific calibration guidelines.

6.5 Concluding Remarks

This thesis has advanced the discriminative drift detection paradigm through four specific methodological contributions: the Prediction Uncertainty Index for entropy-based early warning, adaptive sequential permutation testing for computationally efficient

significance calibration, SHAP interaction-aware drift localisation for correlated feature diagnosis, and the Drift Severity Index for continuous, theoretically motivated severity quantification. The experimental evaluation across synthetic and real-world datasets against seven state-of-the-art baselines confirmed that EADD is the only detector combining complete drift coverage (4/4 types), near-zero false alarms (≤ 1 on drift data, 3.2 mean on stable data), and feature-level attribution with perfect accuracy ($P@k = R@k = F1@k = 1.0$). By answering “Is drift emerging?”, “Is it statistically significant?”, “What caused it?”, and “How severe is it?”, EADD transforms drift detection from a passive binary alarm into an active diagnostic system with severity-proportional automated prescriptions for production MLOps environments.

APPENDIX

APPENDIX A

IMPLEMENTATION DETAILS

To ensure reproducibility, the specific hyperparameters and algorithmic configurations used for the EADD framework are detailed below.

A.1 Adversarial Classifier (LightGBM)

The adversarial classifier utilizes LightGBM (Ke et al., 2017), a high-performance gradient boosting framework, with the following configuration:

Table A.1: LightGBM Hyperparameters

Parameter	Value	Boosting Type
Number of Estimators	50	Learning Rate
Max Depth	0.1	Num Leaves
Objective	31	Binary Classification Metric
Verbosity	AUC	-1

A.2 Windowing and Monitoring

- Reference Window (W_{ref}): 500 samples, maintained via Reservoir Sampling (Vitter, 1985).
- Current Window (W_{cur}): 200 samples, sliding window.
- Monitoring Frequency: Every 50 samples.

A.3 Uncertainty and Statistical Testing

- Prediction Uncertainty Index (PUI): Entropy-based warning.
- Adaptive Sequential Permutation Test (ASPT): Max 50 perms, $\alpha = 0.01$.

A.4 Drift Severity Index (DSI) Tiers

- extbfCritical ($DSI > 0.6$): concentrated univariate.
- extbfModerate ($DSI 0.3-0.6$): subset drift.
- extbfLow ($DSI < 0.3$): diffuse shift.

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BIOGRAPHY

NAME	Ms. Nusrat Begum
DATE OF BIRTH	21 February 1998
PLACE OF BIRTH	Chattogram, Bangladesh
INSTITUTIONS ATTENDED	Stamford International University, 2017–2022 Bachelor of Business Administration (International Business) Mahidol University, 2024–2026 Master of Science (Computer Science) Mahidol University
E-MAIL	nusrat.beg@student.mahidol.edu

